INVESTIGATION OF GROWTH AND CHARACTERIZATION OF A NOVEL 2-[3-(4-HYDROXY-3-METHOXYPHENYL) PROPYL]-1,1,2-TRIMETHYL-1H-BENZO[E] INDOLIUM IODIDE (V-BI) ORGANIC SINGLE CRYSTAL FOR NLO AND TERAHERTZ APPLICATIONS

Dissertation submitted to Periyar University, Salem in partial fulfilment of the requirements for the award of the Degree of

MASTER OF SCIENCE IN PHYSICS

Submitted by

MALLESH J

(Register No: U24PG521PHY016)

Under the Guidance of

Dr. J. KALYANA SUNDAR, M.Sc., Ph.D.,

DEPARTMENT OF PHYSICS

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PERIYAR PALKALAI NAGAR

SALEM - 636 011

APRIL – 2026

Dr. J. KALYANA SUNDAR, M.Sc., M.Phil., Ph.D.,
ASSISTANT PROFESSOR
DEPARTMENT OF PHYSICS
PERIYAR UNIVERSITY
SALEM - 636 011

CERTIFICATE

This is to certify that the project entitled **"INVESTIGATION OF GROWTH AND CHARACTERIZATION OF A NOVEL 2-[3-(4-HYDROXY-3-METHOXYPHENYL) PROPYL]-1,1,2-TRIMETHYL-1H-BENZO[E] INDOLIUM IODIDE (V-BI) ORGANIC SINGLE CRYSTAL FOR NLO AND TERAHERTZ APPLICATIONS"** submitted by **MALLESH J** to Periyar University, Salem in partial fulfillment for the award of the degree of **MASTER OF SCIENCE IN PHYSICS** is a bonafide research work done by the candidate under my supervision during 2024-2026 in the Department of Physics, Periyar University, Salem. The project has not formed the basis for the award of any other Degree and represents independent work on the part of the candidate under my guidance.

Date :

Place : Salem-11

Research Supervisor

(**Dr. J. Kalyana Sundar**)

Head of the Department

DECLARATION

I, **MALLESH J** hereby declare that the project entitled **"INVESTIGATION OF GROWTH AND CHARACTERIZATION OF A NOVEL 2-[3-(4-HYDROXY-3-METHOXYPHENYL) PROPYL]-1,1,2-TRIMETHYL-1H-BENZO[E] INDOLIUM IODIDE (V-BI) ORGANIC SINGLE CRYSTAL FOR NLO AND TERAHERTZ APPLICATIONS"** submitted to Periyar University, Salem in partial fulfilment of the requirements for the award of the **MASTER OF SCIENCE IN PHYSICS** is a bonafide research work carried out by me under the guidance and supervision of **Dr. J. KALYANA SUNDAR., M.Sc., M.Phil., Ph.D.,** Assistant Professor, Department of Physics, Periyar University, Salem and no part of the project has been submitted for award of any other Degree.

Date :

Place : Salem -11

Signature of the Candidate

(**MALLESH J**)

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Abstract

Organic nonlinear optical (NLO) materials have attracted significant attention due to their potential applications in photonic and optoelectronic devices. In the present study, a novel 2-[(4-hydroxy-3-methoxyphenyl) vinyl]-1,1,3-trimethyl-1h-benzo(e)indol-3-ium iodide (V-BI) was successfully synthesized and grown as high-quality single crystals using the slow evaporation solution growth technique. The obtained crystals were subjected to detailed structural, optical, thermal, and nonlinear optical investigations to evaluate their suitability for NLO applications. Single crystal X-ray diffraction analysis confirms the crystalline nature and molecular packing of the compound, revealing a non-centrosymmetric arrangement favourable for nonlinear optical activity. The presence of an extended π-conjugated system within the benzo[e]indole framework enhances intramolecular charge transfer, which plays a crucial role in improving optical nonlinearity. Functional group confirmation was carried out using FTIR spectroscopy. Optical transparency studies carried out using UV–Visible spectroscopy indicate a wide transmission window in the visible region with a suitable optical band gap, ensuring minimal absorption losses for photonic applications. The laser damage threshold (LDT) of the sample was determined, demonstrating the crystal's high tolerance against intense laser irradiation, which is essential for high-power laser applications. Additionally, the second harmonic generation (SHG) efficiency of the grown crystal was evaluated using the Kurtz–Perry powder technique. The combined structural stability, thermal robustness, optical transparency, high laser damage threshold, and enhanced nonlinear response suggest that the 2-[(4-hydroxy-3-methoxyphenyl) vinyl]-1,1,3-trimethyl-1h-benzo(e)indol-3-ium iodide (V-BI) single crystal is a promising candidate for advanced nonlinear optical and frequency conversion applications in modern optoelectronic systems.

Keywords: Nonlinear optics, V-BI, XRD, UV-Vis, FTIR, SHG.

CHAPTER - I
INTRODUCTION OF CRYSTAL GROWTH

1.1 Introduction

Materials science is profoundly driven by the discovery and development of new materials, among which single crystals play a pivotal role [3]. Single crystals are considered the fundamental building blocks for modern technology, and their growth and characterization form the backbone of modern scientific developments in solid-state technology. A single crystal, or monocrystalline solid, is defined as a material in which the crystal lattice of the entire sample is continuous and unbroken to the edges, possessing no grain boundaries.

The absence of defects associated with grain boundaries endows single crystals with unique mechanical, optical, and electrical properties that are often superior to those of polycrystalline materials. Because of their continuous, uniform, and highly-ordered structure, they exhibit intrinsic properties that cannot be replicated in polycrystals. Furthermore, the growth of high-quality single crystals is absolutely necessary to elucidate the intrinsic electronic, magnetic, and thermodynamic properties of materials. Consequently, single crystals have found extensive use in critical technological sectors, including optical, electronic, and optoelectronic applications. For emerging fields like terahertz (THz) technology, the structural perfection of a single crystal is essential to maximize optical transmittance and minimize scattering losses.

1.2 Crystal Growth

Crystal growth is fundamentally a major stage of a crystallization process that consists of the addition of new atoms, ions, or molecular strings into the characteristic arrangement of a crystalline lattice. The action of crystal growth yields a solid whose constituent atoms or molecules are close-packed, maintaining fixed positions in space relative to each other. This transition into the solid, crystalline state can be realized from various parent phases, including the vapor phase, liquid solutions, melts, or even a polycrystalline solid phase.

Thermodynamically, the formation of crystals is classified as a first-order phase transition [4]. This transition is characterized by a non-zero latent heat, known as the crystallization enthalpy, and a discontinuity of the concentration at the phase boundary. The driving force for this transformation is the reduction of free energy; crystallization occurs

because the free energy of the initial solvated or molten state is greater than the free energy of the resulting crystal lattice.

Fig 1.1 Single Crystals

The overall process occurs in two primary stages: nucleation and subsequent growth. Nucleation is the initial stage occurring in the liquid or solution phase, resulting in the formation of small molecular clusters, known as nuclei or embryos. This initial stage requires the system to reach a supersaturated state, making supersaturation the essential driving force for the entire process. After the successful formation of a stable nucleus, the growth stage ensues, during which free particles continuously adsorb onto the nucleus, propagating its highly ordered crystalline structure outwards.

1.3 Classification of crystal growth

The synthesis of single crystals can be achieved through various experimental techniques, which are primarily classified based on the nature of the phase transition involved in the formation of the solid lattice. The fundamental categories include solid-to-solid, liquid-to-solid, and gas-to-solid phase transitions.

Solid-solid growth, such as strain-annealing, involves the rearrangement of grain boundaries within a polycrystalline matrix to form a single crystal, though it is less common for producing large, high-quality bulk optical crystals. Gas-solid or vapor phase growth, which includes techniques like physical vapor deposition (PVD) and chemical vapor deposition (CVD), is highly effective for growing thin films and specific bulk materials that sublimate easily.

The most widely utilized category is liquid-solid growth, which is further subdivided into two major branches: melt growth and solution growth. Melt growth techniques, such as the Czochralski and Bridgman-Stockbarger methods, are predominantly used for materials that melt congruently without decomposition and do not undergo destructive phase transitions upon cooling. However, many advanced optical materials, particularly those being explored for terahertz generation and non-linear optics, possess complex molecular structures that decompose at high temperatures or exhibit severe thermal shock fragility.

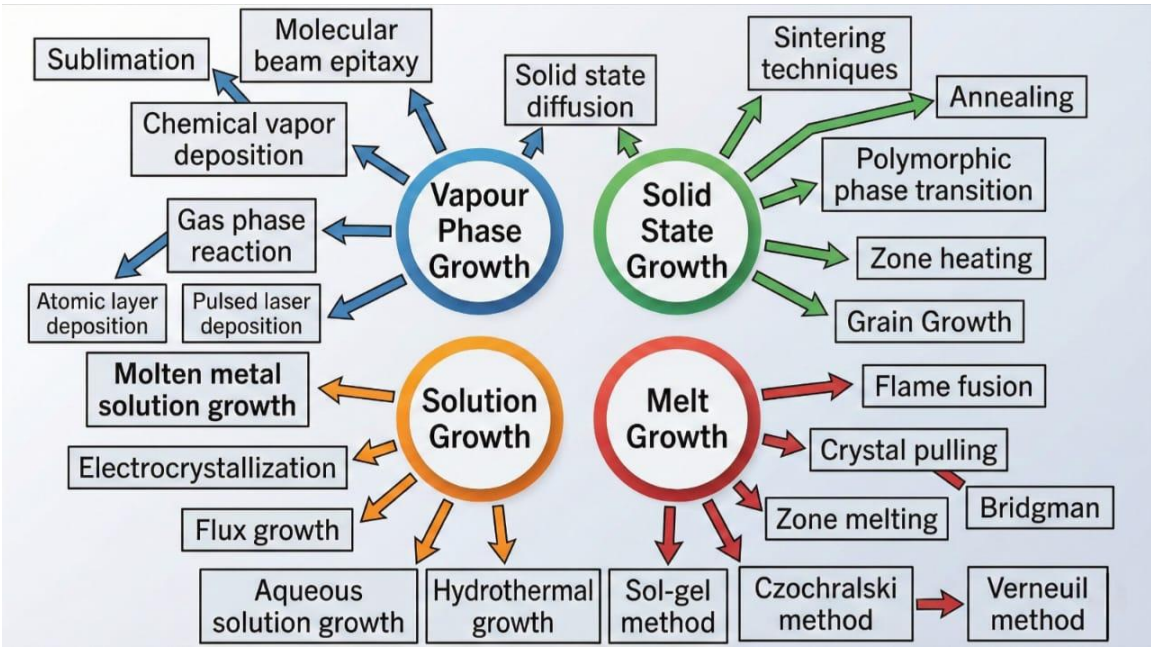

Fig 1.2 Different types of Crystal Growth methods

For such temperature-sensitive materials, solution growth techniques are the preferred alternative [5]. Solution growth occurs at temperatures well below the melting point of the solute, thereby avoiding thermal decomposition and thermally induced strain. This branch includes aqueous and non-aqueous low-temperature solution growth, hydrothermal growth, gel growth, and flux growth. By carefully controlling the thermodynamic parameters of the solvent and solute interactions, solution growth allows for the cultivation of highly perfect, defect-free single crystals suitable for sophisticated optoelectronic applications.

1.4 Processes of crystal growth

The crystallization process from a liquid phase is a complex thermodynamic event that can be distinctly divided into three fundamental and sequential stages: achieving supersaturation, nucleation, and crystal growth.

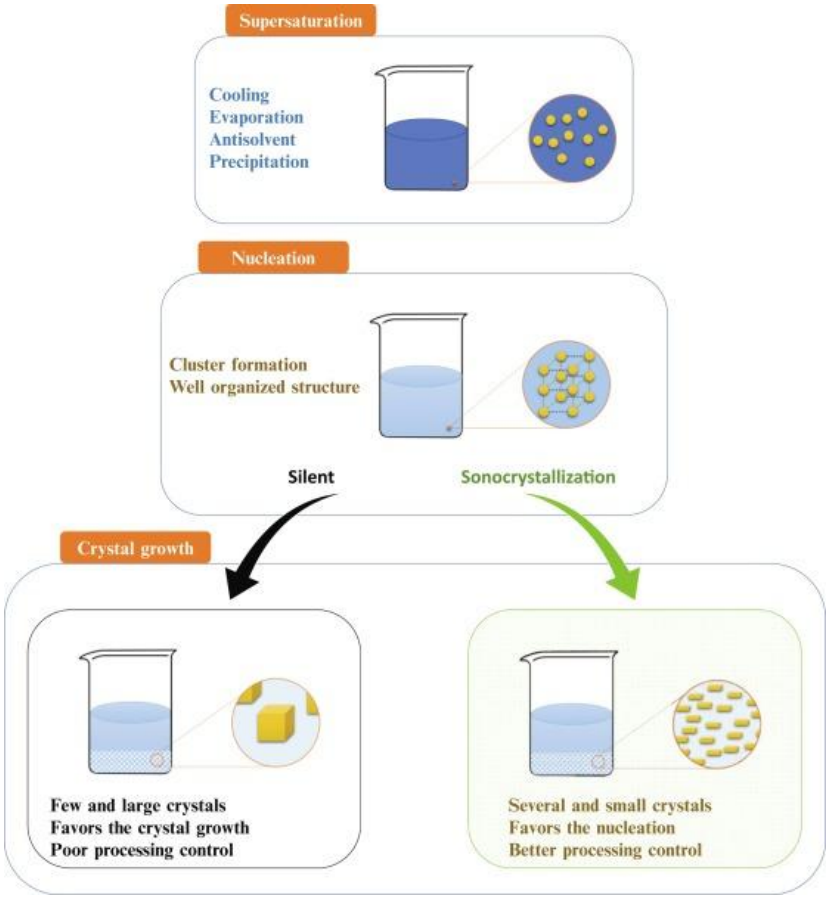

Fig 1.3 Processes of Crystal Growth

The first essential requirement is the creation of a supersaturated state. Supersaturation represents a non-equilibrium condition where the solvent contains a higher concentration of solute than it can thermodynamically support at a given temperature and pressure. This state acts as the primary driving force for crystallization. In solution growth, supersaturation can be achieved through various methods, such as slowly lowering the temperature of the solution, allowing the solvent to evaporate gradually, or introducing a secondary solvent that reduces the overall solubility.

Once adequate supersaturation is reached, the system undergoes nucleation, which is the birth of a new crystalline phase. Nucleation can be either homogeneous or heterogeneous. Homogeneous nucleation occurs spontaneously within the bulk of the solution when solute molecules aggregate to form stable clusters that exceed a critical thermodynamic radius. Heterogeneous nucleation, which requires a lower activation energy, occurs on pre-existing surfaces, such as the walls of the growth vessel, impurities in the solution, or intentionally introduced seed crystals. For the growth of large, single-domain bulk crystals, spontaneous

nucleation is usually suppressed in favour of controlled heterogeneous nucleation on a high-quality seed.

The final stage is the actual crystal growth, which involves the continuous and orderly deposition of solute units onto the stable nucleus or seed. This is a dynamic, multi-step process governed by mass transport and surface integration kinetics. Solute molecules must first diffuse through the bulk solution and the boundary layer to reach the crystal-solution interface. Upon reaching the surface, they must shed their solvation shells, adsorb onto the crystal face, and migrate to energetically favourable sites, typically steps and kinks on the crystal surface, where they are seamlessly incorporated into the advancing crystal lattice. The relative rates of diffusion and surface integration dictate the final morphology and crystalline quality of the grown material.

1.5 Solubility

Solubility is the fundamental thermodynamic parameter that dictates the feasibility, method, and success of any solution-based crystal growth process. It is formally defined as the maximum amount of a solute (the solid material) that can be dissolved in a specific volume of a solvent at a given temperature and pressure to form a thermodynamically stable, homogeneous system. In the context of crystal growth, understanding the solubility behaviour of the synthesized material in various solvents is the critical first step before any growth experiment can begin.

Solutions can exist in three distinct states based on their solute concentration: unsaturated, saturated, and supersaturated. An unsaturated solution can still dissolve more solute, while a saturated solution is in a state of dynamic equilibrium where the rate of dissolution equals the rate of precipitation. Crystal growth, however, strictly requires a supersaturated state, where the solution temporarily holds more solute than the equilibrium value dictates.

The relationship between solubility and temperature is typically expressed through a solubility curve. For most solid materials, solubility exhibits a positive temperature coefficient, meaning that the substance becomes more soluble as the temperature of the solvent increases. The exact steepness or slope of this solubility curve determines which specific solution growth technique, such as slow evaporation or slow cooling, will be most effective for yielding high-quality single crystals for optoelectronic and terahertz applications.

1.5.1 Metastable Zone Width (MSZW)

The metastable zone width (MSZW) is a fundamental thermodynamic and kinetic parameter that must be optimized to achieve successful bulk single crystal growth from a solution. It is defined graphically on a concentration-temperature plot as the region bounded by the equilibrium solubility curve (where the solution is exactly saturated) and the supersolubility curve (the point at which spontaneous primary nucleation occurs).

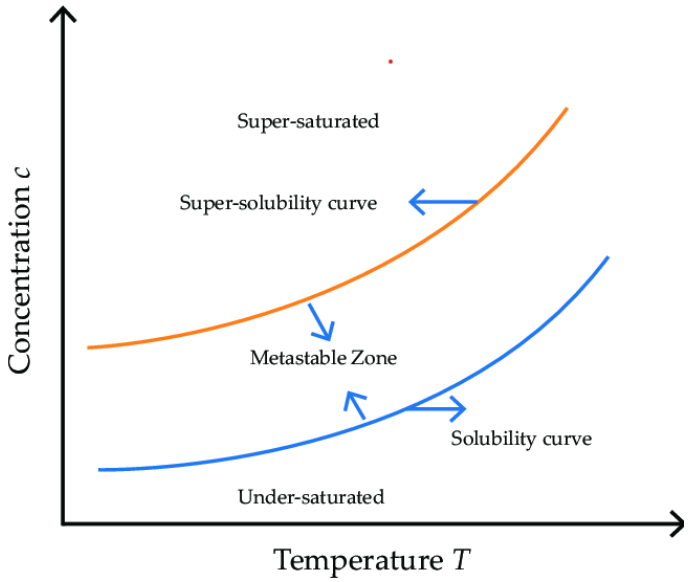

Fig 1.4 Metastable Zone in C vs T graph

Understanding and controlling the MSZW is arguably the most critical operational aspect of the solution growth method. If the supersaturation level of a solution exceeds the upper limit of the metastable zone, the system enters the labile region, resulting in massive, spontaneous nucleation [6]. This leads to the rapid formation of multiple tiny crystallites, severely depleting the solute and completely ruining the chances of growing a single, large crystal.

Conversely, maintaining the solution's concentration strictly within the boundaries of the metastable zone ensures a highly stable supersaturated environment. In this zone, spontaneous homogeneous nucleation is thermodynamically suppressed. If a high-quality seed crystal is introduced into this metastable solution, the excess solute will preferentially deposit onto the seed rather than forming new nuclei. Therefore, materials that exhibit a uniquely wide metastable zone width are highly prized in crystal growth, as they permit higher growth rates and greater stability against minor temperature fluctuations, ultimately yielding the large, defect-free single crystals required for sensitive optical and terahertz devices.

1.6 Solution Growth Techniques

1.6.1. High Temperature Growth Techniques

1.6.1.1 Hydrothermal growth

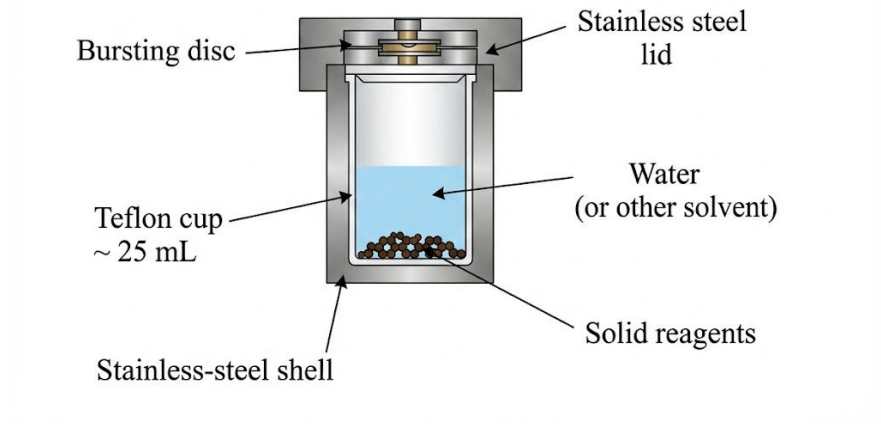

Fig 1.5 Schematic of a Hydrothermal experiment

Hydrothermal growth is a specialized and highly effective technique for synthesizing single crystals that are otherwise difficult to grow due to their insolubility in water or other solvents under ambient conditions. The term "hydrothermal" refers to the action of water at elevated temperatures and pressures, typically above the normal boiling point of water and occasionally exceeding its critical point (374°C and 22.1 MPa). Under these extreme conditions, the physical and chemical properties of water, such as its dielectric constant, density, and viscosity change significantly, greatly enhancing its solvent power for many inorganic substances, including silicates, oxides, and complex salts [7].

The growth process is carried out in a thick-walled steel pressure vessel known as an autoclave, which is designed to withstand highly corrosive environments and extreme internal pressures. The process relies on the establishment of a temperature gradient within the autoclave. Nutrient material (powder or small crystals) is placed in the hotter, lower zone of the vessel, while seed crystals are suspended in the cooler, upper zone. As the solvent is heated, it dissolves the nutrient at the bottom. Due to the temperature gradient, convection currents transport the saturated solution upward to the cooler region. Here, the solution becomes supersaturated, and the dissolved material precipitates onto the seed crystals, resulting in continuous, high-quality crystal growth. This method is particularly valued for producing large, defect-free crystals like synthetic quartz, which are essential for precise optical and piezoelectric applications.

1.6.1.2 Flux growth Method

While the hydrothermal method is a highly pressurized high-temperature technique typically utilized for growing inorganic oxides and quartz, the reflux method is an unpressurized high-temperature solution technique that is absolutely indispensable for the synthesis of complex organic optical materials. It is primarily employed to drive endothermic chemical reactions that require sustained, elevated thermal energy over extended periods without suffering the continuous loss of volatile solvents.

The fundamental physical principle of the reflux method relies on the continuous cycling of solvent phases. The experimental apparatus typically consists of a reaction vessel—such as a round-bottom flask containing the chemical precursors and a specific solvent—securely fitted with a vertically oriented, water-cooled Liebig or Allihn condenser. The reaction vessel is placed on a heating mantle or an oil bath and brought to the boiling point of the chosen solvent.

As the solution boils, the solvent vaporizes and travels upward into the vertical condenser. The continuous flow of cold water through the condenser's outer jacket rapidly extracts thermal energy from the hot vapor, causing it to undergo an immediate phase change back into a liquid state. This condensed liquid then continuously drips back down into the boiling reaction mixture.

This continuous boiling and condensation cycle provides several critical advantages for crystal engineering:

1. **Isothermal Stability:** It safely maintains the reaction at a precise, constant temperature (the exact boiling point of the solvent) for extended durations—often ranging from several hours to multiple days, without requiring complex electronic temperature controllers.

2. **Solvent Preservation:** It completely prevents the evaporation and depletion of the reaction solvent, ensuring the molar concentrations of the reactants remain perfectly stable throughout the synthesis.

3. **Synthesis of Organic NLO Precursors:** In the development of organic non-linear optical (NLO) and terahertz crystals, the reflux method is the standard technique used to synthesize the initial molecular compounds. The sustained thermal energy ensures complete thermodynamic conversion and high-yield synthesis of highly conjugated, polar organic molecules, such as donor-π-acceptor styryl dyes, which are subsequently

isolated, purified, and grown into macroscopic single crystals using lower-temperature techniques like the slow evaporation method.

1.6.2. Low-Temperature Solution Growth

Low-temperature solution growth is widely recognized as one of the most versatile and effective techniques for cultivating highly perfect, large-dimension single crystals, particularly for materials that cannot withstand the extreme thermal stresses of melt growth. Operating typically at temperatures below 100°C, this method prevents the thermal decomposition of complex molecular structures, limits the formation of thermally induced strain, and drastically reduces the concentration of structural defects like vacancies and dislocations [43]. This structural perfection is paramount for crystals intended for terahertz applications, where defects can cause severe scattering and optical loss.

The process relies on the slow and controlled precipitation of the solute from a supersaturated solution.

1.6.2.1 Slow cooling and Slow Evaporation method

The two most prominent methods within this category are the slow cooling method and the slow evaporation method. The choice between the two depends heavily on the material's solubility curve. The slow cooling technique is employed when a material exhibits a steep positive solubility curve; as the temperature of a saturated solution is precisely and gradually lowered, the solution becomes supersaturated, forcing the excess solute to crystallize onto a seed.

Conversely, the slow evaporation technique is the method of choice for materials with a relatively flat solubility curve, where solubility does not change significantly with temperature. In this approach, the solution is maintained at a constant temperature, and supersaturation is driven entirely by the gradual evaporation of the solvent. By carefully controlling the evaporation rate, often by restricting the atmospheric exposure of the growth vessel, the solute molecules slowly and meticulously integrate into the crystalline lattice. This controlled environment promotes exceptional optical transparency and crystal face development [8].

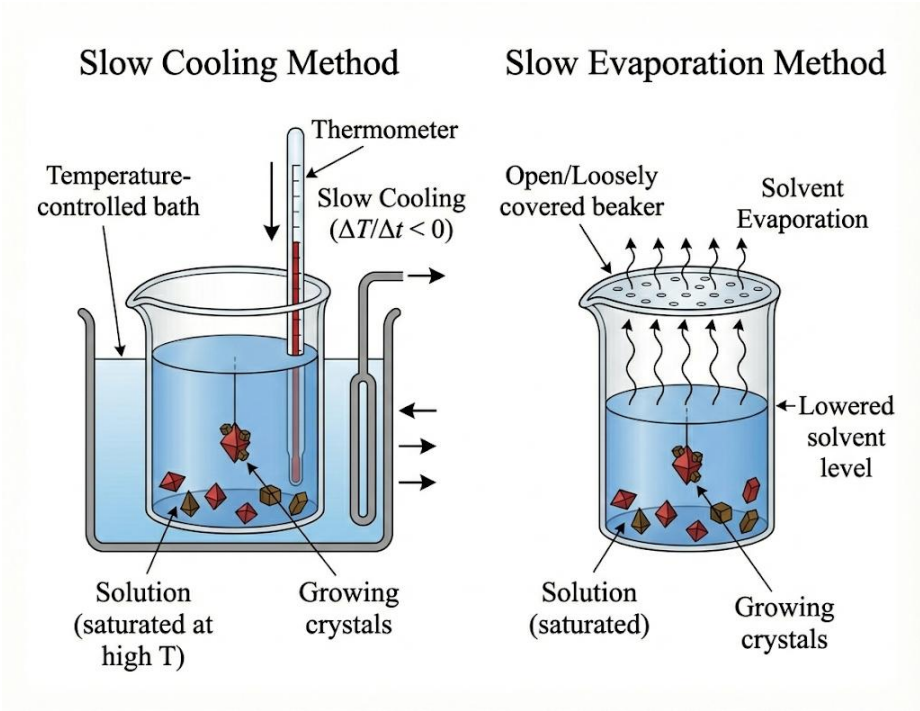

Fig 1.6 Schematic of Slow cooling and Slow Evaporation method

1.7 Non-Linear Optics (NLO)

In conventional linear optics, when a low-intensity electromagnetic wave (such as light from a standard bulb) interacts with a dielectric medium, the induced macroscopic dielectric polarization (P) is directly and linearly proportional to the applied electric field (E). This relationship is defined as:

$$P = \epsilon_0 \chi^{(1)} E$$

Where ϵ_0 is the vacuum permittivity and $\chi^{(1)}$ is the linear optical susceptibility of the medium. However, with the advent of high-power laser systems, the electric field of the incident light becomes comparable to the internal interatomic electric fields of the material. Under these intense irradiation conditions, the linear approximation collapses, and the induced polarization responds non-linearly. The total polarization must then be expressed as a Taylor series expansion of the applied electric field [9]:

$$P = \epsilon_0 \left(\chi^{(1)} E + \chi^{(2)} E^2 + \chi^{(3)} E^3 + \cdots \right) = P^{(1)} + P^{(NL)}$$

Here, $\chi^{(2)} and \chi^{(3)}$ represent the second-order and third-order non-linear optical susceptibilities, respectively. These higher-order terms ($P^{(NL)}$) are responsible for a wide array of non-linear frequency conversion phenomena.

1.7.1 Second-Order Non-Linear Effects

Second-order non-linear effects arise from the $P^{(2)} = \epsilon_0 \chi^{(2)} E^2$ term in the polarization expansion. If we consider an incident monochromatic laser beam with an electric field $E(t) = E_0 \cos(\omega t)$, substituting this into the second-order term yields:

$$P^{(2)}(t) = \epsilon_0 \chi^{(2)} E_0^2 \cos^2(\omega t) = \frac{1}{2} \epsilon_0 \chi^{(2)} E_0^2 [1 + \cos(2\omega t)]$$

This mathematically reveals two distinct and highly important physical phenomena:

1. **Second Harmonic Generation (SHG):** The $\cos(2\omega t)$ term dictates that the material acts as an oscillating dipole emitting electromagnetic radiation at exactly **twice** the frequency (2ω) of the incident light. Physically, two low-energy photons (ω) are destroyed to simultaneously create one high-energy photon (2ω).

2. **Optical Rectification (OR):** The constant, frequency-independent term ($\frac{1}{2} \epsilon_0 \chi^{(2)} E_0^2$) represents a static, DC polarization field generated across the crystal. When an ultrashort (femtosecond) laser pulse is used instead of a continuous wave, this "DC" term essentially follows the intensity envelope of the laser pulse, emitting an electromagnetic transient. This specific Optical Rectification process is the fundamental physical mechanism used to generate broad-bandwidth Terahertz (THz) radiation [44].

Symmetry Requirement: It is a strict rule of quantum mechanics and crystallography that the $\chi^{(2)}$ tensor is exactly zero in any material possessing a center of inversion. Therefore, second-order effects like SHG and Terahertz generation via OR can **only** occur in non-centrosymmetric crystals (such as the $P2_1$ space group of the synthesized V-BI crystal).

1.8 Introduction of Non-Linear Optical crystals

Nonlinear optics (NLO) is a specialized branch of modern optical physics that explores the behavior of light in non-linear media-materials where the dielectric polarization responds nonlinearly to the electric field of the incident light. In classical, linear optics, when low-intensity light passes through a dielectric medium, the induced polarization of the electron clouds is directly proportional to the applied oscillating electric field. This linear relationship governs familiar optical phenomena such as refraction, reflection, and standard absorption.

However, the advent of high-intensity light sources, specifically lasers, revolutionized this field. When a material is illuminated by a laser, the strength of the external electric field becomes comparable to the internal interatomic electric fields binding the electrons to their nuclei. Under such intense electromagnetic excitation, the linear approximation fails. The material's polarization response becomes anharmonic, generating higher-order polarization waves that act as new sources of electromagnetic radiation [11].

This nonlinear interaction is the cornerstone of modern laser technology and optoelectronics. It allows materials to alter the frequency, phase, or other characteristics of the incident light, enabling phenomena such as second harmonic generation (frequency doubling), sum-frequency generation, and optical rectification. Notably, an absolute structural prerequisite for a crystal to exhibit these second-order nonlinear optical effects is that its crystal lattice must lack a center of inversion symmetry, categorizing it as a non-centrosymmetric crystal. The development of new non-centrosymmetric single crystals is currently one of the most active areas of materials science, particularly for extending these nonlinear frequency conversion techniques down into the terahertz regime.

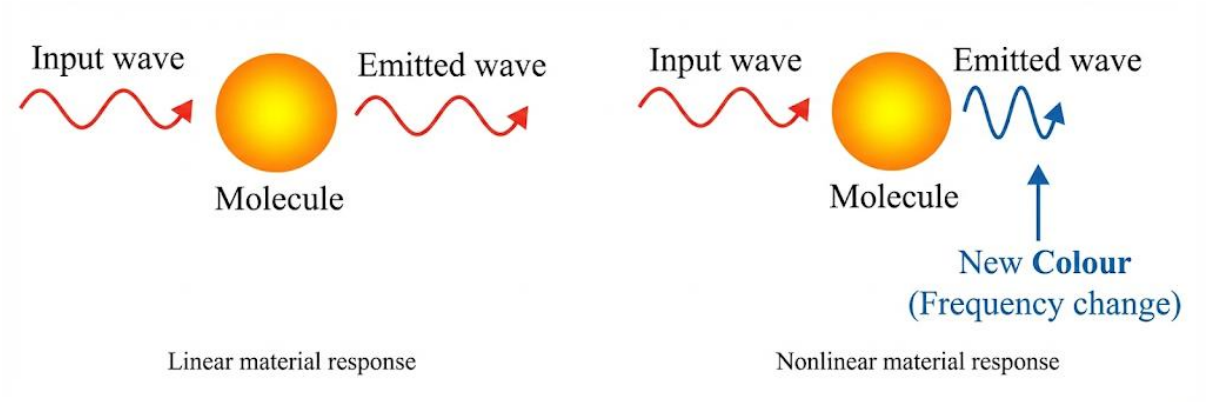

Fig 1.7 Response of Linear and Non-linear materials

1.9 Properties of Non-Linear materials

For a single crystal to be effectively utilized in sophisticated nonlinear optical devices, particularly for the generation and detection of terahertz waves, it must satisfy a highly stringent set of optical, thermal, and mechanical criteria. The absolute fundamental requirement for any second-order nonlinear effect to occur is that the material must crystallize in a non-centrosymmetric space group. The lack of an inversion center in the crystal lattice ensures that the material possesses a non-zero second-order non-linear susceptibility tensor. Beyond this structural prerequisite, an ideal non-linear material must inherently possess a large non-linear

optical coefficient, which directly dictates the energy conversion efficiency from the incident pump laser to the generated terahertz frequency.

Equally critical is the crystal's optical transparency profile. A high-quality non-linear crystal must exhibit a broad optical transmission window that simultaneously accommodates the wavelength of the pump laser, typically in the visible or near-infrared spectrum, and the subsequently generated terahertz radiation. Any significant intrinsic absorption in either of these operational regimes will severely attenuate the output signal and rapidly generate localized heat [12]. Consequently, the material must also demonstrate a remarkably high laser damage threshold. Because non-linear frequency conversion processes are typically inefficient and require excitation by high-peak-power pulsed lasers, the crystal lattice must be structurally robust enough to withstand intense electromagnetic radiation without suffering optical breakdown, localized melting, or point defect generation. Furthermore, appropriate phase-matching capabilities, usually achieved through the material's natural birefringence, are essential to ensure that the generated optical waves constructively interfere and amplify throughout the entire interaction length of the crystal.

1.10 Introduction to Terahertz (THz) Radiation

Terahertz (THz) radiation refers to the region of the electromagnetic spectrum that lies strategically between the macroscopic world of high-frequency microwave electronics and the microscopic quantum world of infrared optics. Typically defined as encompassing frequencies from 0.1 to 10 THz (where 1 THz = 10^{12} Hz). For decades, this region was notoriously referred to as the "THz gap" [13]. This gap existed because traditional electronic devices (like transistors and antennas) are limited by carrier transit times and cannot operate fast enough to generate these frequencies, while traditional photonic devices (like semiconductor lasers) are limited by the thermal energy at room temperature, which swamps the low-energy transitions required to emit THz photons.

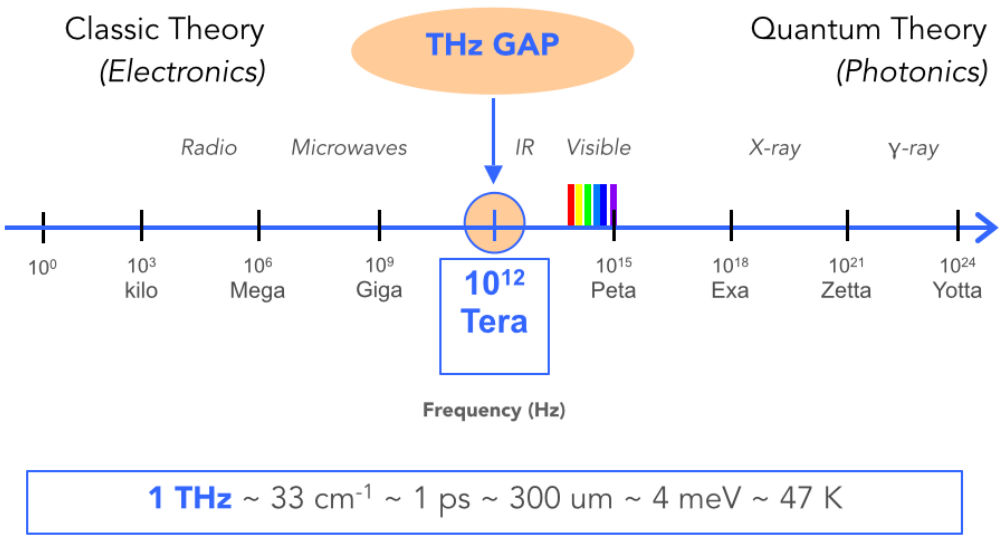

Fig 1.8 Terahertz Gap in Electromagnetic setup

In recent years, advancements in materials science and non-linear optics have successfully bridged this gap, unlocking the immense potential of THz radiation. THz waves possess several unique and highly advantageous properties. Unlike X-rays, THz radiation is non-ionizing because its photon energy is extremely low (approximately 4.1 meV at 1 THz, making it inherently safe for biological tissues and medical imaging [14]. Furthermore, THz waves are highly transparent to many common non-polar dielectric materials, such as plastics, paper, ceramics, and clothing, allowing for non-destructive testing and security screening.

Crucially for scientific research, many complex molecules, including pharmaceuticals, explosives, and biological macromolecules, exhibit strong, distinct rotational and vibrational transition resonances within the THz band. This makes terahertz spectroscopy a powerful tool for chemical fingerprinting. However, the widespread commercialization of THz technology is heavily reliant on the development of efficient, low-cost, and robust solid-state THz sources and detectors. High-quality, non-centrosymmetric single crystals, grown carefully from solution, are currently at the forefront of this technological push, serving as the essential non-linear media for converting standard laser light into coherent terahertz radiation.

1.11 Objective of the Present Work

The primary aim of this research is the successful synthesis, growth, and comprehensive characterization of a novel non-linear optical single crystal intended for terahertz applications. Specifically, this project focuses on utilizing low-temperature solution growth techniques to

obtain a structurally perfect crystal and subsequently investigating its physical, structural, and optical properties.

The systematic objectives of the present investigation are defined as follows:

- To successfully synthesize the high-purity starting material through controlled chemical reactions.
- To optimize the low-temperature solution growth parameters (such as temperature, evaporation rate) to suppress secondary nucleation and grow a visually transparent, defect-free, single crystal.
- To confirm the crystalline perfection, determine the lattice parameters, and identify the crystal system using X-ray Diffraction (XRD) analysis.
- To identify the presence of specific functional groups and confirm the molecular structure of the grown crystal using Fourier Transform Infrared (FTIR) spectroscopy.
- To analyze the optical transmission window, determine the lower cut-off wavelength, and calculate the optical bandgap of the crystal using UV-Vis spectroscopy, ensuring it meets the transparency requirements for non-linear optical applications.
- To determine the structural resilience of the crystal lattice under high-power ultrafast laser irradiation by conducting a single-shot Laser Damage Threshold analysis.
- To experimentally quantify the macroscopic second-order non-linear optical conversion efficiency by evaluating its Second Harmonic Generation (SHG) output relative to standard reference materials using the Kurtz and Perry powder technique.

CHAPTER - II
CHARACTERIZATION TECHNIQUES

2.1 Introduction

The successful growth of a single crystal is only the first step in materials science research. Once a bulk crystal is synthesized and harvested from the solution, it is imperative to rigorously evaluate its physical, structural, optical, and thermal properties to ascertain its suitability for the intended application in this case, terahertz (THz) generation and non-linear optics (NLO). Characterization techniques act as the analytical bridge between the experimental synthesis and the practical device application.

A single analytical tool is insufficient to provide a complete profile of a newly grown crystal. Therefore, a systematic series of characterization techniques must be employed. Structural characterizations, primarily utilizing X-ray diffraction, are required to confirm the crystalline phase, determine the unit cell dimensions, and verify the absence of structural phase impurities. Spectroscopic techniques, spanning from the infrared to the ultraviolet regions, are essential to identify molecular functional groups, confirm the molecular structure, and determine the optical transmission window and bandgap. Furthermore, thermal analysis is necessary to establish the material's melting point and thermal stability under high-power laser irradiation. Finally, specialized NLO and THz time-domain spectroscopies evaluate the actual functional performance of the crystal. This chapter details the fundamental physical principles and operational theories underlying each of the analytical instruments used in this present investigation.

2.2 X-Ray Diffraction analysis

X-ray diffraction (XRD) is the most definitive, non-destructive analytical technique utilized in solid-state physics for determining the atomic and molecular structure of a crystal. To investigate the internal arrangement of atoms within a solid, the probing radiation must possess a wavelength that is on the same order of magnitude as the interatomic distances within the crystal lattice. Visible light, with wavelengths ranging from 400 to 700 nm, is vastly too large to resolve atomic structures. X-rays, however, are high-energy electromagnetic radiation with wavelengths typically ranging from 0.01 to 10 nm (often expressed in Angstroms, where 1 Å = 0.1 nm. Because the typical interplanar spacing in a crystal lattice is roughly 1 to 3 A, the crystal acts as an ideal naturally occurring three-dimensional diffraction grating for X-rays.

When a monochromatic beam of X-rays strikes a crystalline material, the incident photons interact primarily with the electron clouds of the constituent atoms. The atoms scatter the incident X-rays in all directions, a process known as elastic or Rayleigh scattering. In most directions, these scattered waves are out of phase and destructively interfere, canceling each other out. However, in specific, highly defined directions, the scattered waves are perfectly in phase and constructively interfere, producing a strong diffracted beam.

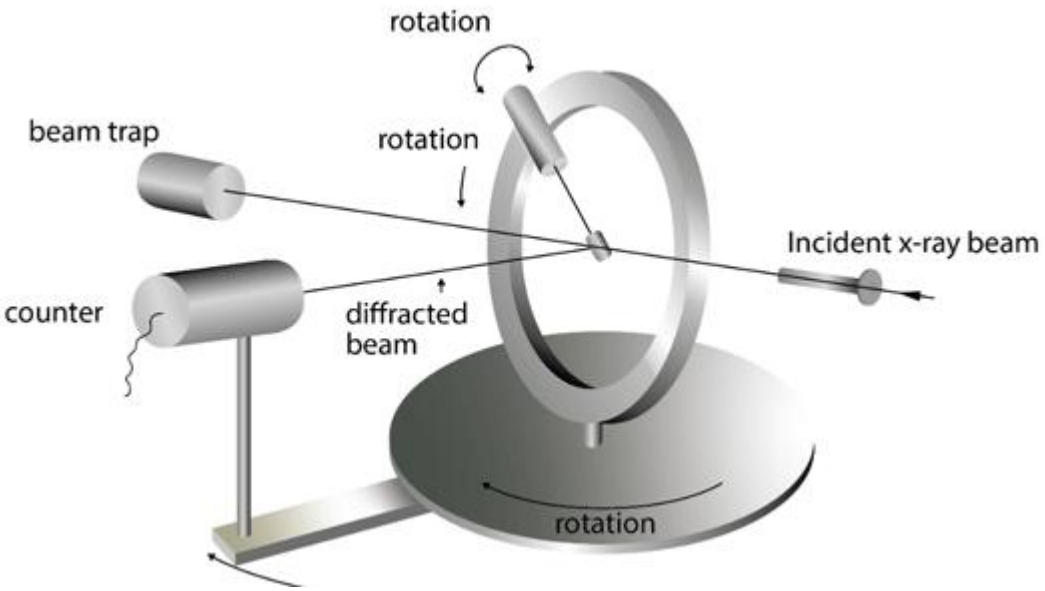

Fig. 2.1 Illustration of Single-crystal X-ray diffraction

The geometric conditions required for this constructive interference were formulated by W.L. Bragg and W.H. Bragg in 1913. They modeled the crystal lattice as a series of parallel atomic planes separated by a constant distance, d. For constructive interference to occur, the path length difference between two X-ray beams reflecting off adjacent parallel planes must be an integer multiple of the incident X-ray wavelength. This fundamental relationship is expressed mathematically by Bragg's Law [26]:

$$n\lambda = 2dsin\theta$$

Where:

- n is an integer representing the order of diffraction (n = 1, 2, 3,...).
- λ is the wavelength of the incident monochromatic X-ray beam (typically 1.5406 A for a Copper Kα source).
- d is the interplanar spacing between the parallel atomic planes in the crystal lattice.

- θ is the glancing angle of incidence (the Bragg angle) at which the maximum diffraction intensity occurs.

By experimentally measuring the specific angles θ at which constructive interference produces a diffraction peak and knowing the wavelength (λ) of the X-ray source, Bragg's Law can be rearranged to calculate the exact interplanar spacing (d) of the atomic planes. These d-spacing values serve as a unique structural fingerprint for the material, allowing researchers to unequivocally identify the crystalline phase and deduce the lattice parameters (a, b, c, α, β , γ) of the newly grown crystal.

2.3 Powder X-ray Diffraction

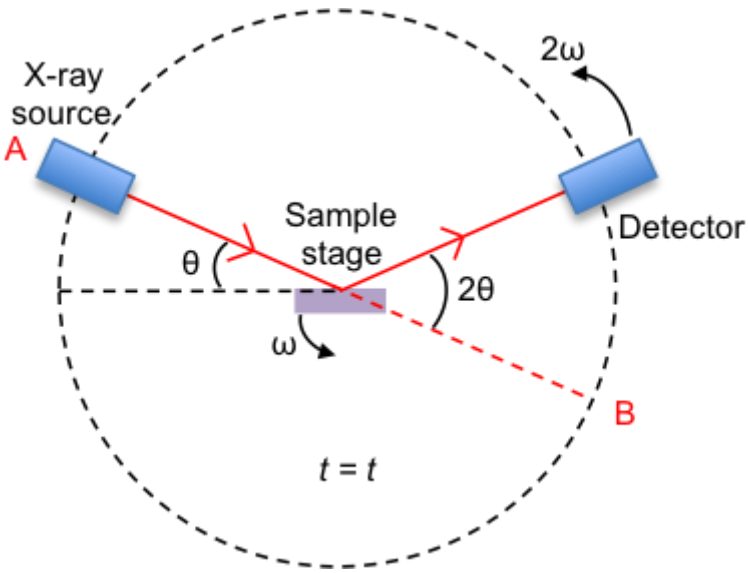

Fig. 2.2 Illustration of Powder X-ray diffraction

While single-crystal X-ray diffraction is used to solve the complete three-dimensional atomic structure of an unknown material, Powder X-ray Diffraction (PXRD) is the standard and most efficient technique for phase identification, structural confirmation, and assessing the crystalline purity of a synthesized material. To prepare a sample for PXRD, a small portion of the grown single crystal is mechanically ground into a fine, homogeneous powder. This powder consists of millions of randomly oriented microscopic crystallites.

Because these crystallites are randomly oriented in every possible geometric direction, all possible crystallographic planes (hkl) are simultaneously exposed to the incident X-ray beam. Consequently, for any given set of atomic planes with a specific interplanar spacing (d), there will always be a statistically significant fraction of crystallites oriented at the exact Bragg angle (θ) required to satisfy Bragg's Law and produce constructive interference. Instead of the

distinct diffraction spots produced by a single crystal, the diffracted beams from a powder sample form continuous concentric cone of radiation, known as Debye-Scherrer cones, emanating from the sample.

In a modern powder diffractometer, a detector systematically sweeps across a range of 2θ angles, recording the intensity of the diffracted X-rays. The resulting data is plotted as a diffractogram, which displays the diffraction intensity as a function of the diffraction angle (2θ). The positions of these peaks are intrinsically linked to the unit cell dimensions (lattice parameters) of the crystal, while the relative intensities of the peaks are determined by the types of atoms present and their specific coordinates within the unit cell. By comparing the obtained diffractogram against a standard reference database, such as the Joint Committee on Powder Diffraction Standards (JCPDS), the crystalline phase and purity of the grown material can be unequivocally confirmed.

Additionally, the average crystallite size (D) of the powdered sample can be estimated from the broadening of the most intense diffraction peak using the prominent Scherrer equation [27]:

$$D = \frac{K\lambda}{\beta cos\theta}$$

Where K is the dimensionless shape factor (typically roughly 0.9), λ is the X-ray wavelength, β is the full width at half maximum (FWHM) of the diffraction peak measured in radians, and θ is the Bragg angle.

2.4 Fourier transform infrared spectral analysis

Fourier Transform Infrared (FTIR) spectroscopy is a fundamental, non-destructive characterization tool utilized to investigate the vibrational and rotational modes of molecules. It is primarily employed to identify specific functional groups, characterize covalent bonds, and confirm the molecular structure of the synthesized crystalline material.

The physical principle of FTIR relies on the interaction between electromagnetic radiation in the infrared region (typically 4000 to 400 cm^{-1}) and the chemical bonds within the sample. Molecules are not static; their constituent atoms are constantly vibrating. When the sample is irradiated with a broad-spectrum infrared beam, the chemical bonds will absorb energy at specific resonant frequencies that perfectly match the natural frequency of their vibrational modes, such as stretching, bending, or rocking. However, a strict quantum

mechanical selection rule dictates that for a molecular vibration to be "IR active" and absorb infrared radiation, the vibrational motion must result in a net change in the permanent electrical dipole moment of the molecule [28].

Unlike older dispersive spectrometers that scan through individual frequencies one by one, an FTIR spectrometer measures all infrared frequencies simultaneously. This is achieved using a Michelson interferometer, which splits the incident infrared beam, reflects it off a fixed mirror and a moving mirror, and recombines the beams to create an interference pattern known as an interferogram. This interferogram is a complex time-domain signal containing all the spectral information of the sample. To extract the useful frequency-domain spectrum (a plot of transmittance or absorbance versus wavenumber), a mathematical operation called a Fourier Transform is applied to the raw data by a computer.

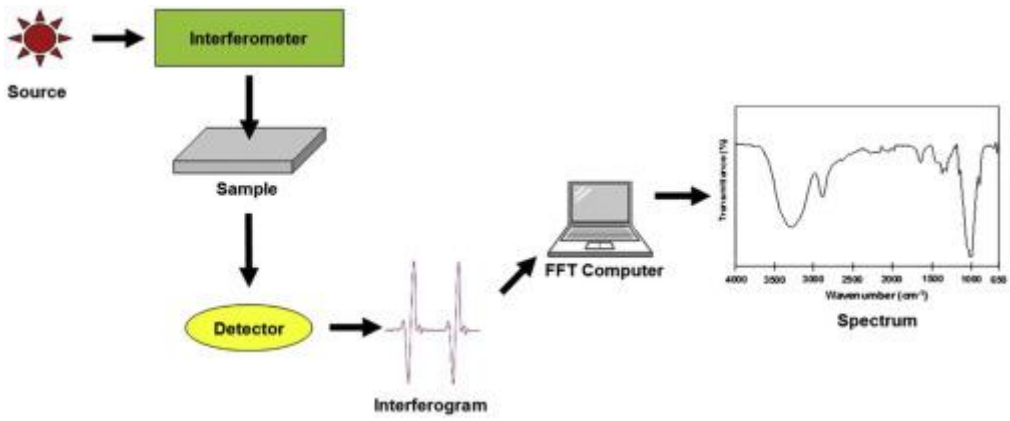

Fig 2.3 Illustration of FTIR Spectroscopy

For the development of crystals intended for terahertz and non-linear optical applications, FTIR analysis is exceptionally critical. Not only does it confirm that the correct molecular structure has been synthesized, but it also detects the presence of unwanted impurities. For instance, broad absorption bands in the higher wavenumber region often indicate the presence of trapped water molecules or residual solvent within the crystal lattice, which can severely diminish the material's optical transparency and terahertz generation efficiency.

2.5 UV-Visible spectral analysis

Ultraviolet-Visible-Near Infrared (UV-Vis-NIR) spectroscopy is a vital characterization technique used to quantify the optical properties of a synthesized crystal. While FTIR probes the vibrational states of molecules, UV-Vis-NIR spectroscopy investigates the fundamental

electronic transitions within the atoms or molecules. When a crystal is irradiated with light in the wavelength range of roughly 190 nm to 1100 nm, photons with sufficient energy are absorbed, exciting valence electrons from their ground state (typically bonding or non-bonding orbitals) to higher-energy anti-bonding orbitals.

For materials intended for non-linear optical (NLO) and terahertz applications, optical transparency is arguably the most critical evaluation parameter. The crystal must possess a broad, uninterrupted transmission window that encompasses the wavelength of the excitation pump laser (typically in the visible or near-infrared region). Any intrinsic absorption at the pump wavelength will lead to a drastic reduction in the efficiency of the non-linear frequency conversion process, while simultaneously causing localized heating and subsequent laser-induced damage to the crystal lattice. UV-Vis-NIR spectral data is recorded as either absorbance or transmittance as a function of wavelength. A high-quality NLO crystal will exhibit a remarkably high and flat transmittance profile across its operational range, terminating abruptly at its lower cut-off wavelength, which corresponds to the fundamental absorption edge where electrons are excited directly across the bandgap [29].

Furthermore, the optical absorption data obtained from the spectrometer is mathematically manipulated to determine the optical bandgap (E_g) of the material using the Tauc relation:

$$\alpha h\nu = A(h\nu - E_g)^n$$

Where:

- $\alpha$is the linear absorption coefficient.
- $h\nu$ is the incident photon energy.
- A is an energy-independent constant.
- n is an index that characterizes the specific nature of the electronic transition (e.g., n = 1/2 for direct allowed transitions, and n = 2 for indirect allowed transitions).

By plotting $(\alpha h\nu)^{1/n}$ against the photon energy $h\nu$ (a graph widely known as a Tauc plot) and extrapolating the linear linear portion of the curve to the energy axis (where α= 0), the precise optical bandgap of the newly grown crystal can be accurately calculated.

2.6 Laser Damage Threshold (LDT) Analysis

For single crystals intended for application in non-linear optics (NLO) and terahertz (THz) wave generation, optical transparency and high conversion efficiency are not the only critical parameters. The material must also exhibit a remarkably high tolerance against intense electromagnetic irradiation. Because non-linear frequency conversion processes are typically inefficient, they demand excitation from high-peak-power, ultrafast pulsed lasers. Consequently, determining the Laser Damage Threshold (LDT) is a mandatory characterization step to ensure the physical, structural, and thermodynamic robustness of the grown crystal before its integration into commercial optoelectronic devices.

Laser-induced optical damage in organic dielectric crystals typically originates from two primary physical mechanisms: localized thermal damage and avalanche breakdown. Thermal damage arises when intrinsic structural defects, trapped solvent inclusions, or specific molecular chromophores linearly absorb the fundamental laser energy. This localized absorption results in rapid heating, localized melting, and eventual thermal shock fracturing of the lattice. Alternatively, avalanche breakdown occurs under the extreme electric fields of the laser pulse, where multi-photon absorption excites and accelerates free electrons, creating a dense, localized plasma that violently ruptures the macroscopic crystal structure.

In a standard LDT experimental setup, the crystal is subjected to high-energy pulses generated by a Q-switched Nd:YAG laser, typically operating at a fundamental near-infrared wavelength of $1064\ \mathrm{nm}$. The laser beam is routed through a variable attenuator to strictly control the input pulse energy and is subsequently directed through a converging biconvex lens to tightly focus the beam onto a precisely cleaved or optically polished plane of the crystal face. The exact incident energy is continuously monitored in real-time using a calibrated optical power and energy meter.

The measurement is systematically performed by gradually increasing the incident laser energy, either on a single site (multiple-shot method) or on fresh macroscopic sites for every pulse (single-shot method), until catastrophic physical damage, such as surface pitting, internal cracking, or a distinct plasma spark, is visually or microscopically observed. The specific energy (E) at which this dielectric breakdown occurs is recorded as the threshold energy. The laser damage threshold power density (P_d), conventionally expressed in $\mathrm{GW/cm^2}$, is then quantitatively calculated using the fundamental relation:

$$P_d = \frac{E}{\tau \pi r^2}$$

Where E represents the threshold energy in Joules, τ is the full-width at half-maximum (FWHM) pulse duration of the laser in seconds, and r is the exact radius of the focused laser spot on the crystal surface. Materials possessing a high calculated LDT demonstrate robust intermolecular packing and low intrinsic linear absorption, making them exceptionally viable for high-power laser applications [31].

2.7 Second Harmonic Generation (SHG) Efficiency

To experimentally confirm that a newly synthesized crystalline material belongs to a non-centrosymmetric space group and possesses the inherent capability for second-order non-linear optical interactions, Second Harmonic Generation (SHG) testing is performed. The most widely adopted preliminary diagnostic tool for evaluating the SHG efficiency of new materials is the Kurtz and Perry powder technique, developed in 1968 [30]. This technique is highly favored because it provides a rapid, reliable assessment of the material's non-linear optical phase-matching capabilities without the immediate prerequisite of growing a massive, perfectly oriented, optical-quality single crystal.

In the standard Kurtz-Perry experimental setup, a small portion of the grown crystal is mechanically ground into a fine powder and densely packed into a micro-capillary tube. This powder sample is then illuminated by a high-intensity, Q-switched Nd:YAG (Neodymium-doped Yttrium Aluminum Garnet) fundamental pulsed laser, typically operating at a fundamental wavelength of 1064 nm in the near-infrared region.

If the crystalline lattice of the material strictly lacks a center of inversion symmetry, the intense electromagnetic field of the incident 1064 nm photons will induce a non-linear polarization within the dielectric medium. Through the second-order non-linear susceptibility tensor, two incident low-energy photons are annihilated to simultaneously generate a single, high-energy photon at exactly twice the fundamental frequency (and half the wavelength). Therefore, the successful generation of a bright, visible green light emission at 532 nm directly from the powdered sample serves as unequivocal proof of second harmonic generation.

The optical signal of the generated green light is passed through an infrared-blocking filter, collected by a photomultiplier tube, and displayed on an oscilloscope. By comparing the millivolt amplitude of this output signal to that of a well-known standard reference material of similar particle size, such as microcrystalline Potassium Dihydrogen Phosphate (KDP) or Urea,

the relative SHG conversion efficiency of the newly grown crystal can be quantitatively determined.

CHAPTER - III
LITERATURE SURVEY

Recent advancements in benzoindolium-based styryl dyes, such as P-BI, have demonstrated highly efficient NLO and terahertz characteristics [1]. A comprehensive review published in the Journal of Electromagnetic Waves and Applications detailed the fundamental boundaries of the THz gap, defining its typical frequency and wavelength spectrum [15]. Building upon this, research in Trends in Biotechnology highlighted the biomedical applications of THz spectroscopy and imaging, emphasizing the non-ionizing nature and low photon energy of THz waves, which make them uniquely suited for biological applications [16]. To generate these waves, studies in Advanced Materials established the stringent requirements for new nonlinear optical (NLO) materials, emphasizing the critical importance of phase matching, high laser damage thresholds, and broad transparency windows [17]. The specific mechanism of transferring near-infrared laser pulses into THz radiation utilizing second-order NLO effects via optical rectification was systematically detailed in the IEEE Transactions on Terahertz Science and Technology [18].

In the domain of organic NLO materials, studies in the Journal of Materials Chemistry C discussed the pivotal role of extended π-conjugation and benzo-fused indoles in significantly enhancing intramolecular charge transfer [19]. Research published in Dyes and Pigments detailed the synthesis and optical properties of indolium and benzoindolium-based styryl dyes, focusing on the chemical mechanisms of Knoevenagel condensation and the reactivity of the 2-methyl group [20]. Furthermore, literature in Advanced Materials explored the molecular engineering of these materials, highlighting the influence of steric hindrance from gem-dimethyl groups in promoting non-centrosymmetric macroscopic crystal packing [21]. The utilization of asymmetric hydrogen-bonding networks to force this necessary non-centrosymmetric macroscopic crystal packing to achieve high nonlinear coefficients was further established in Chemical Reviews [22].

The thermodynamic principles governing optimal crystal growth have been extensively surveyed in CrystEngComm and the Journal of Crystal Growth, which analyzed the measurement and interpretation of the metastable zone width (MSZW) and its direct impact on single crystal yield and the suppression of secondary nucleation within the Ostwald-Miers diagram [23]. The Handbook of Crystal Growth detailed the specific mechanisms of thermal control and slow evaporation from low-temperature solutions. Additionally, literature in

Materials Science and Engineering provided comprehensive overviews of specialized low-temperature techniques, such as gel growth, emphasizing how gel matrices suppress convection and reduce nucleation sites to yield highly perfect single crystals. The foundational development of organic crystals for THz generation and detection was extensively reviewed by Jazbinsek et al.. [24].

Meier et al. highlighted the high-efficiency THz generation capabilities of similar organic DAST crystals [25]. For the functional evaluation of synthesized materials, S.K. Kurtz and T.T. Perry published a foundational paper in the Journal of Applied Physics detailing a powder technique for the evaluation of nonlinear optical materials, which remains the standard methodology for assessing Second Harmonic Generation (SHG) efficiency. Structural characterization techniques were outlined by texts from Pearson Education and Dover Publications, which provided the fundamental theory of X-ray scattering, Bragg's equation, and the application of the Scherrer equation for crystallite size estimation. Furthermore, the identification of IR active molecular vibrations and functional group frequency assignments using Fourier Transform Infrared (FTIR) spectrometry was comprehensively guided by texts from John Wiley & Sons. Finally, the mathematical derivation of the Tauc relation and the physics of direct and indirect bandgaps for interpreting optical transparency via UV-Vis spectrophotometry were thoroughly established in publications by Oxford University Press. The structural requirements for maximizing macroscopic non-linearities in D-π-A systems were outlined by Marder et al.

Conclusion:

- The unique non-ionizing properties and low photon energy of terahertz waves offer groundbreaking, safe applications in biomedical imaging and non-destructive chemical detection.
- However, practically bridging the "THz gap" requires overcoming the inefficiencies of current technologies by engineering new, low-cost organic materials with massive second-order nonlinear coefficients and high laser damage thresholds.
- Therefore, developing and characterizing a novel terahertz crystal is a critical, high-impact endeavour to provide the highly efficient frequency-conversion media necessary to power the next generation of advanced optoelectronic devices.

CHAPTER - IV
MATERIALS AND METHODS

4.1 Materials and Reagents

The synthesis and subsequent bulk single crystal growth of organic materials for terahertz and non-linear optical applications demand precursor chemicals of exceptional purity. Even trace amounts of elemental impurities can introduce significant structural defects into the crystalline lattice, drastically increase optical absorption losses, and inadvertently act as active centers for unwanted secondary nucleation during the solution growth process. Therefore, all primary organic compounds and solvents utilized in this investigation were of analytical reagent (AR) grade and were used strictly as received, without undergoing any further purification processes.

4.1.1 1,1,2-Trimethyl-1H-benzo[e]indole

1,1,2-Trimethyl-1H-benzo[e]indole (Chemical formula: $C_{15}H_{15}N$) is a polycyclic aromatic heterocyclic compound that serves as the foundational organic precursor for the synthesis of the target terahertz crystal. indole chemical structure.

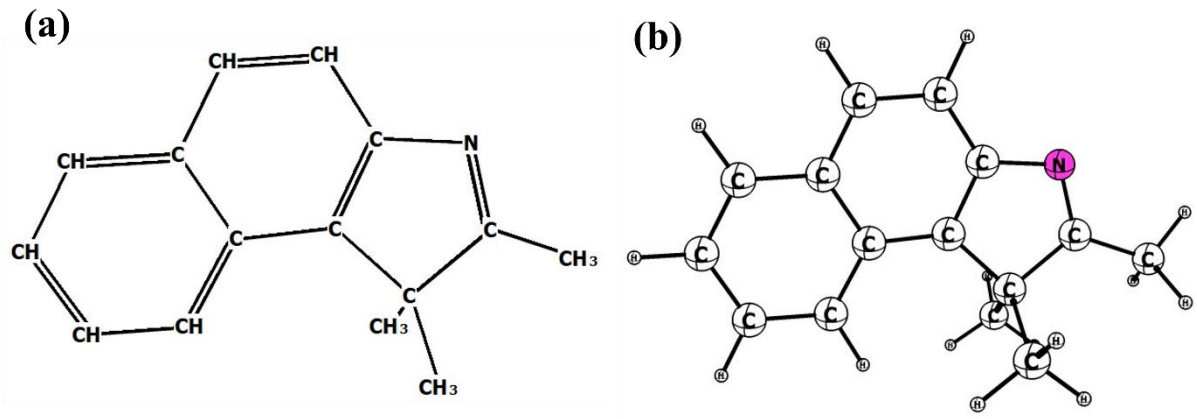

Fig 4.1 (a) 2D representation of 1,1,2-Trimethyl-1H-benzo[e]indole
(b) 3D representation of 1,1,2-Trimethyl-1H-benzo[e]indole

Physical and Chemical Properties:

Structurally, this molecule is characterized by an indole ring system fused to an additional benzene ring, creating a rigid, highly extended naphthalene-like π-conjugated network. The strategic selection of this specific benzo-fused derivative over a standard indole is primarily driven by this extended π-conjugation. In the development of organic non-linear

optical (NLO) materials, an extended and planar π-electron system allows for highly efficient intramolecular charge transfer (ICT) across the molecule when subjected to an external optical electric field [32].

Furthermore, the substitution pattern of this molecule is chemically tailored for the synthesis of advanced optoelectronic chromophores. The gem-dimethyl groups located at the 1-position provide critical steric hindrance. This steric bulk serves a dual purpose: it prevents unwanted side reactions at that carbon center during synthesis, and more importantly, during the subsequent crystal growth phase, it dictates the molecular packing arrangement. For a crystal to generate terahertz radiation, it must crystallize in a non-centrosymmetric space group; the asymmetrical bulk provided by these methyl groups strongly encourages this non-centrosymmetric packing by preventing the molecules from aligning perfectly anti-parallel to one another.

Chemically, the methyl group positioned at the 2-carbon of the pyrrole ring exhibits a high degree of CH-acidity. This reactive active methyl group is the designated reaction site for forming complex styryl dyes or hemicyanine derivatives. It readily undergoes base-catalyzed or acid-catalyzed Knoevenagel condensation reactions [34] with aromatic aldehydes.

Relevance to Terahertz Applications:

In the context of designing high-efficiency non-linear optical crystals for terahertz applications, 1,1,2-Trimethyl-1H-benzo[e]indole acts as a potent electron-donating moiety and a robust π -bridge in a donor- π -acceptor (D- π -A) molecular architecture. When condensed with an electron-accepting or structurally polarizable aldehyde, the resulting molecule exhibits an extreme asymmetry in its electron density. This pronounced asymmetric electron distribution yields an exceptionally large first molecular hyperpolarizability. When these highly polarizable molecules are successfully packed into an acentric macroscopic crystal lattice, they produce a massive second-order non-linear susceptibility(χ^2), which is the strict fundamental requirement for converting ultrashort laser pulses into coherent terahertz radiation via optical rectification.

4.1.2 Vanillin

Vanillin, systematically designated as 4-Hydroxy-3-methoxybenzaldehyde (Chemical formula: $C_8H_8O_3$), is a naturally occurring phenolic aldehyde utilized as the essential co-reactant in the synthesis of the target non-linear optical crystal.

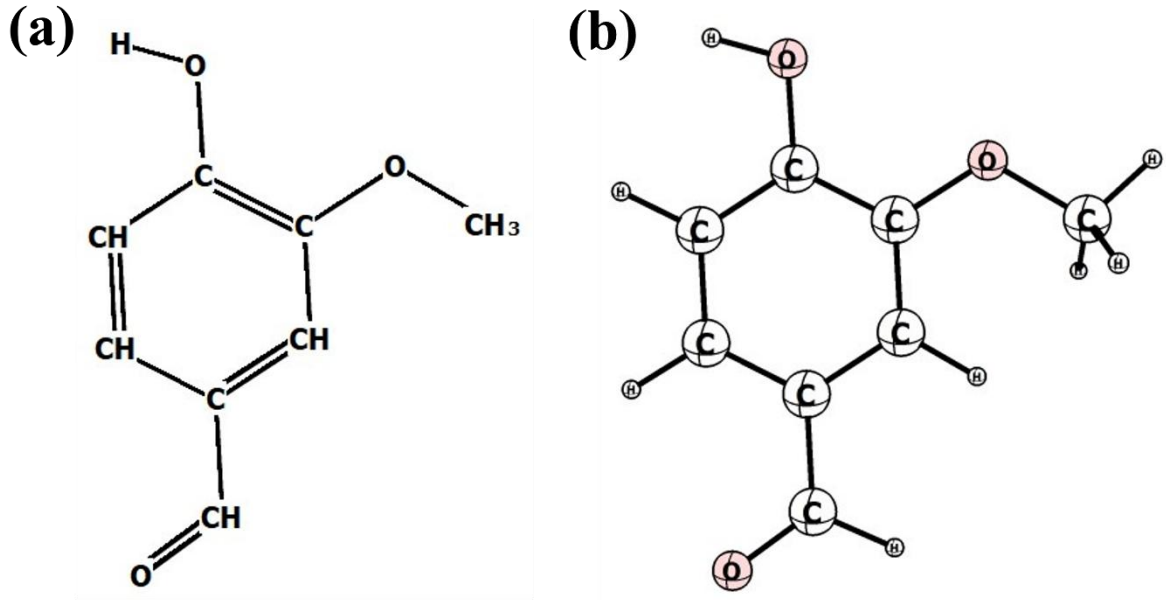

Fig 4.2 (a) 2D representation of 4-Hydroxy-3-methoxybenzaldehyde
(b) 3D representation of 4-Hydroxy-3-methoxybenzaldehyde

Physical and Chemical Properties:

Vanillin is a highly functionalized aromatic compound, featuring an aldehyde group (-CHO), a hydroxyl group (-OH), and a methoxy group ($-OCH_3$) substituted on a central benzene ring. Each of these functional groups plays a distinct, critical role in both the chemical synthesis and the ultimate optical performance of the resulting crystalline material.

Chemically, the highly reactive, electrophilic carbonyl carbon of the aldehyde group is perfectly suited to undergo a condensation reaction (specifically, a Knoevenagel-type condensation) with the nucleophilic active methyl group of the 1,1,2-Trimethyl-1H-benzo[e]indole precursor. This specific chemical coupling creates a rigid, conjugated ethylenic (vinyl) double bond, effectively linking the two aromatic systems together. This newly formed

double bond acts as an extended π-electron bridge, allowing for seamless electronic communication between the two halves of the synthesized styryl dye molecule.

Relevance to Terahertz Applications and Crystal Engineering:

In the architecture of organic non-linear optical (NLO) materials, vanillin contributes significantly to the formation of a highly polarized donor- π -acceptor (D- π -A) system. The hydroxyl (-OH) and methoxy ($-OCH_3$) groups on the vanillin moiety act as powerful electron-donating auxochromes. When coupled across the π -bridge to the electron-withdrawing indolium-based core, the molecule experiences a massive internal dipole moment. Upon excitation by an external pump laser, this structure allows for an ultra-fast, highly efficient unidirectional intramolecular charge transfer (ICT), resulting in an exceptionally large first molecular hyperpolarizability (β).

Beyond molecular hyperpolarizability, vanillin is prized in the field of NLO crystal engineering strictly for its structural asymmetry. One of the greatest challenges in developing organic terahertz crystals is the natural thermodynamic tendency of highly polar D- π -A molecules to crystallize in a head-to-tail, centrosymmetric arrangement (where dipoles cancel each other out, yielding zero macroscopic second-order non-linear effects). The specific 3-methoxy and 4-hydroxy substitution pattern of vanillin actively disrupts this anti-parallel pairing. The hydroxyl group acts as a strong directional hydrogen-bond donor, while the asymmetric steric bulk of the methoxy group forces the molecules into an acentric (non-centrosymmetric) macroscopic crystal lattice. This specific crystal packing is the absolute fundamental prerequisite for the material to exhibit the macroscopic second-order non-linear susceptibility (χ^2) necessary for high-efficiency terahertz wave generation via optical rectification.

4.2 Synthesis Procedure

The synthesis of the target non-linear optical compound, V-BI, was carried out utilizing meticulously cleaned glassware to prevent any trace contamination. All utensils, including round-bottom flasks and beakers, were washed thoroughly and rinsed twice with double-distilled water prior to use. The synthesis followed a systematic two-step reflux condensation route.

4.2.1 Experimental procedure

Synthesis Procedure of 1,1,2,3-tetramethyl-1H-benzo[e]indol-3-ium iodide (BI)

1,1,2,3-tetramethyl-1H-benzo[e]indol-3-ium iodide was prepared by taking an equimolar ratio of 1,1,2-trimethyl-1H-benzo[e]indole (0.627 g) and Methyl iodide (0.425 ml) in 30 ml of Toluene. The mixtures were refluxed and heated at 110 ^{o} C overnight for 16h. After completion of the reflux period, the flask was removed from the heating mantle, covered with an aluminium foil, and allowed to cool gradually at room temperature. The resulting mixture was then filtered utilizing Whatman filter paper and the obtained product was washed with di ethyl ether to remove unreacted starting material. The collected product was transferred to a clean petri dish and placed in a hot air oven at 80 ^{o} C for 2 hours to ensure complete drying, yielding the Product I (BI)

4.2.2 Synthesis Procedure of V-BI

The title material V-BI was prepared by taking stoichiometric ratio of 1,1,2,3-tetramethyl-1H-benzo[e]indol-3-ium iodide and 4-Hydroxy-3-methoxybenzaldehyde (vanillin) in 50 ml of methanal and piperidine was added as a catalyst (few drops). The above mixtures were taken in a 250ml round bottom flask and the mixture solution was refluxed for 18h. After the reflux, the flask was removed, covered with aluminium foil, and cooled for 20 minutes. The resulted red solid was filtered using Whatman filter paper and washed 3-5 times with Diethyl ether to remove unreacted starting materials. The purity of the synthesised compound could be improved by repeated re crystallisation at least thrice using methanol solution.

4.3 Harvesting of Crystal

Single crystals of V-BI were grown by slow evaporation technique at ambient temperature. The Supersaturated solution of V-BI was dissolved in methanol: Acetonitrile (3:1) mixed solvent system. The solution was filtered by using Whatman filter paper. The filtered

solution was kept in 100mL beaker, and it was tightly covered with aluminium foil with few holes for solvent evaporation. After a period of 12 days A transparent good quality crystal has obtained. The grown crystal was shown in Fig 4.4. Visually, the harvested V-BI crystal exhibits a striking deep ruby-red coloration and possesses a distinct prismatic and deeply fluted morphology. The crystal habit forms a prominent elongated prism accompanied by parallel, stepped growth patterns along its primary faces. In terms of optical clarity, the crystal is sub-transparent to translucent, while the thickest portions of the bulk heavily absorb light due to the concentrated dye structure, the thinner facets and sharp edges allow light to penetrate, revealing a clean, homogenous internal red structure. Macroscopic physical inspection confirms that the principal crystal faces are highly reflective, smooth, and possess sharply defined, clean-cut edges. The complete absence of visible opaque inclusions, solvent bubbles, or internal macroscopic cracks indicates a high degree of internal structural perfection. This combination of robust prismatic growth and sub-transparent clarity visually validates the excellent crystalline quality of the synthesized V-BI material, confirming its suitability for subsequent non-linear optical (NLO) and optoelectronic characterizations.

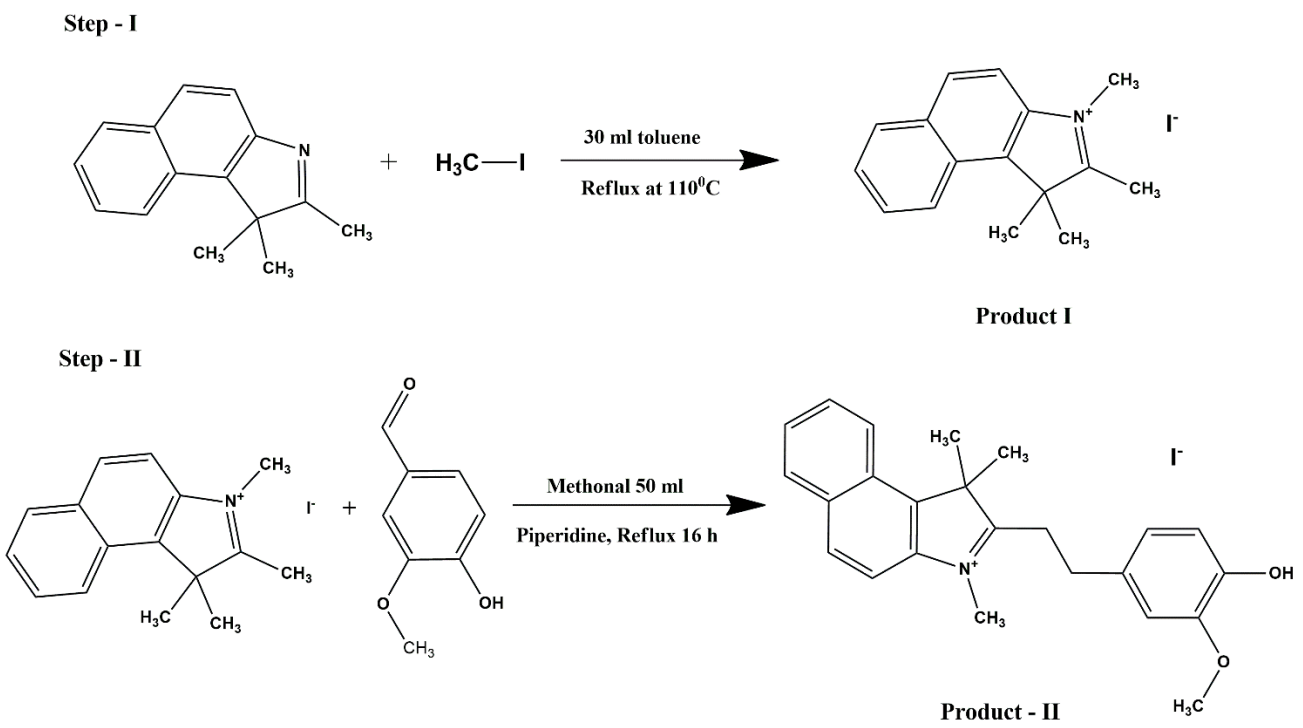

Fig 4.3 Synthesis scheme of the V-BI sample

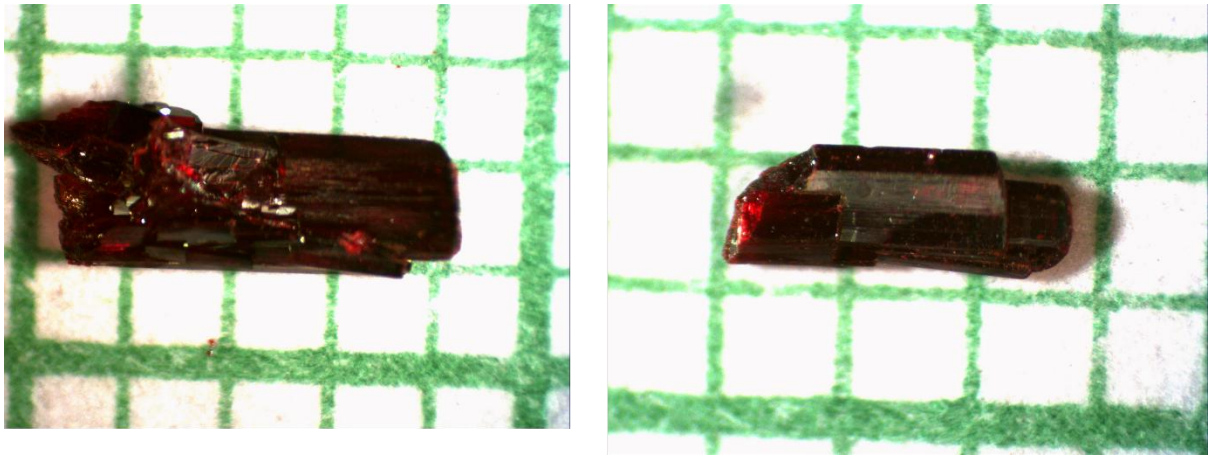

Fig 4.4. Grown crystals

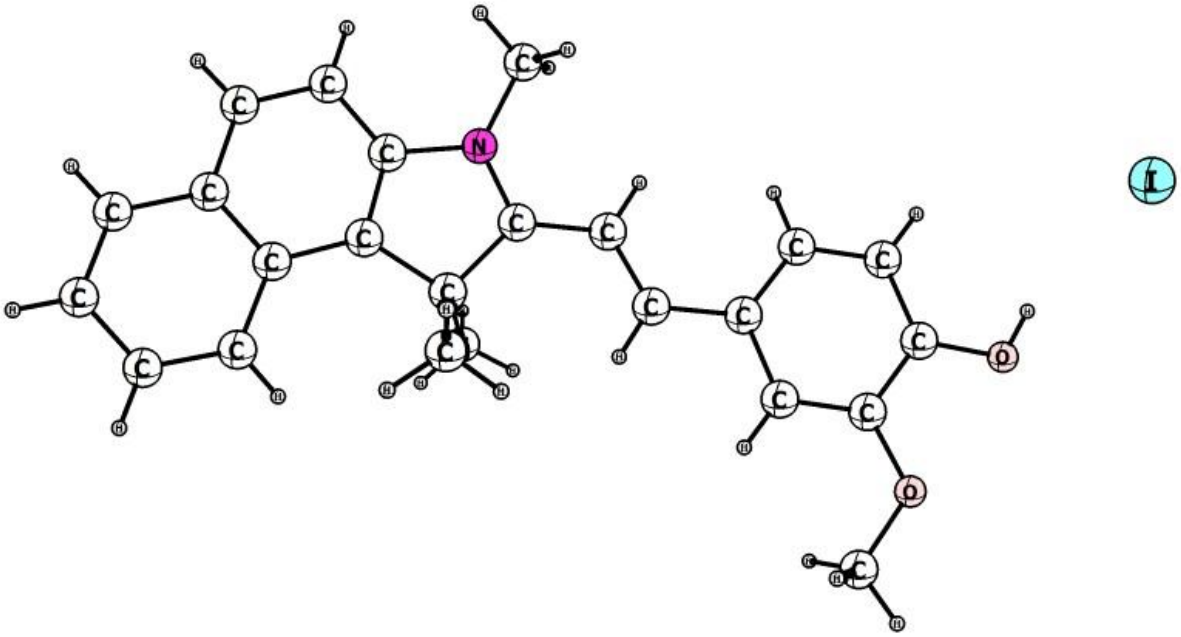

Fig 4.5 Molecular structure of synthesised V-BI

CHAPTER V

RESULTS AND DISCUSSION

5.1 Introduction

The successful synthesis and subsequent crystallization of the target styryl compound, V-BI, represent the critical experimental phase of this investigation. Following the harvest of the needle-like single crystals, grown via the constant-temperature slow evaporation technique from the optimized mixed solvent system, it is imperative to rigorously evaluate their structural, molecular, and optical properties. This chapter systematically presents and interprets the analytical data obtained from a specific suite of characterization techniques to ascertain the crystal's suitability for advanced optical applications.

5.2 X-ray Diffraction (XRD) Analysis

The structural parameters of the V-BI grown crystal were determined using a Bruker D8 Quest single-crystal X-ray diffractometer. From the single-crystal X-ray diffraction, the obtained unit cell parameter values confirmed that the crystal belongs to the monoclinic crystal system with the strictly non-centrosymmetric $P2_1$ space group. The accuracy of the solved structure is highly publishable, yielding an exceptionally low final R-factor of 0.0332 (3.32%), a Goodness-of-Fit (S) value of 1.076, and a Flack parameter of 0.07(2), strictly satisfying standard crystallographic publication criteria. The structural parameters and crystallographic data of the grown crystal are summarized in Table 5.1.

To establish the structural novelty and distinct packing arrangement of the V-BI single crystal, its refined unit cell parameters were compared with previously reported analogy structures, namely P-BI [35] and C-BI [36] (Table 5.1). While all three crystals belong to the monoclinic crystal system, V-BI crystallizes in the strictly non-centrosymmetric $P2_1$ space group, similar to P-BI, which is the fundamental prerequisite for second-order non-linear optical and terahertz wave generation activity. In contrast, the C-BI crystal crystallizes in the centrosymmetric $P2_1/n$ space group, making it applicable for third-order, rather than second-order, NLO applications. Furthermore, the significant variations in the specific lattice dimensions (a, b, c) and unit cell volume explicitly confirm that V-BI is a newly engineered, structurally distinct organic crystal.

The grown crystal powder sample was also characterized by a powder X-ray diffractometer equipped with a Cu-Kα radiation source. The finely crushed powder sample was scanned, and the obtained powder XRD pattern is presented in Figure 5.2. The experimental powder XRD pattern of the V-BI crystal was compared with the theoretically simulated pattern generated from the solved CIF file using MERCURY software. From the results, the crystallographic data are well matched with the experimental data, which confirms the correct structural formation of the V-BI compound. Based on the observed diffraction pattern of the grown crystal, it can be confirmed that all the observed diffraction lines have well-defined sharp peaks, indicating the good crystalline nature of the titled compound. Moreover, we could not find any additional or extraneous peaks in the observed diffraction patterns, confirming the high phase purity of the grown crystal. Additionally, the average crystallite size of the powdered sample was calculated to be approximately 45.8 nm using the Debye-Scherrer equation [42].

Parameter	**V-BI (Present Work)**	**P-BI [35]**	**C-BI [36]**
Molecular formula	C24 H24 I N O2	C25 H27 I N2	C23 H21 N Cl I
Molecular weight (g/mol)	485.34	482.39	473.76
Space group	$P2_1$	$P2_1$	$P2_1/n$
Crystal system	Monoclinic	Monoclinic	Monoclinic
Unit cell parameters			
a (Å)	9.9596	7.34	7.2849
b (Å)	9.3349	12.59	21.9014
c (Å)	12.2652	12.53	12.4896
α (°)	90°	90°	90°
β (°)	107.033°	105.93°	96.36°
γ (°)	90°	90°	90°
Volume (Å³)	1090.30	1109.00	1992.70

Table 5.1: Structural parameters of V-BI compared with reported analogy

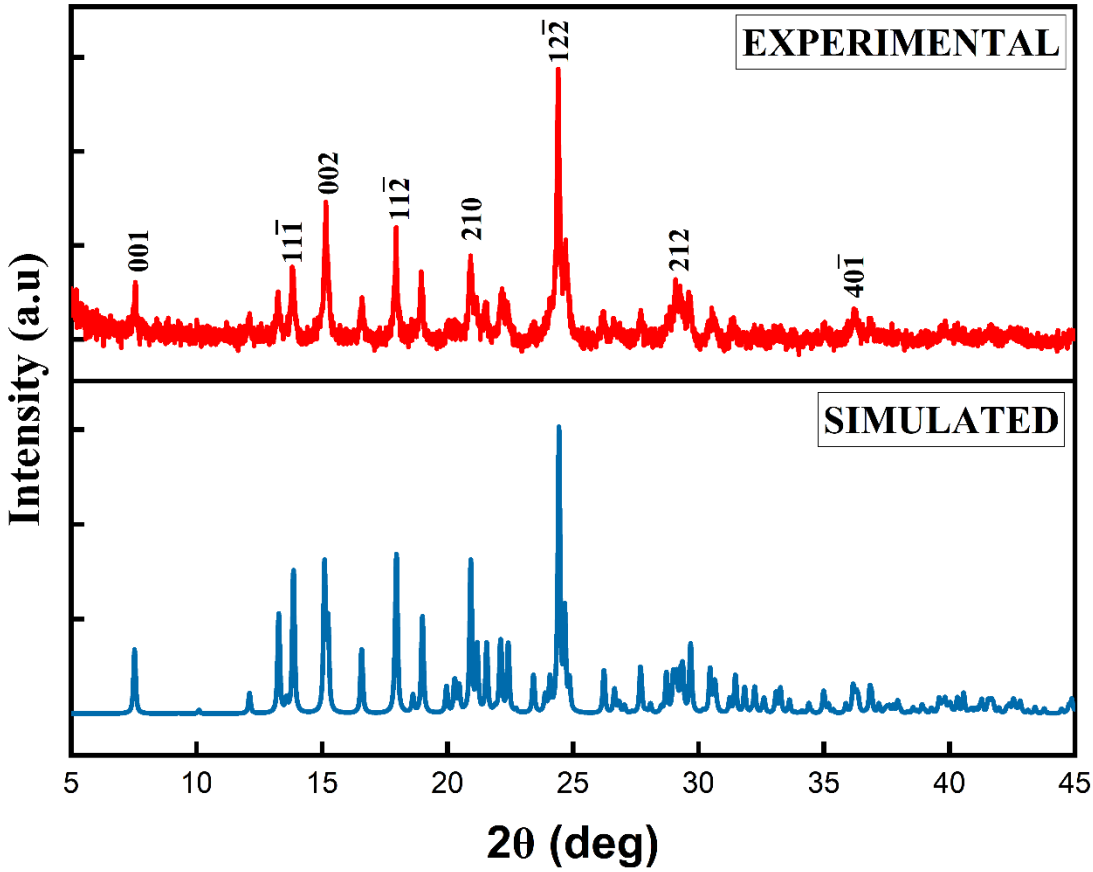

Fig 5.2 Powder X-Ray diffraction pattern of V-BI

5.3 Fourier Transform Infrared (FTIR) Spectral Analysis

Analysis of the FTIR spectrum is a substantial proof that provides more information on a structured compound. A Fourier Transform Infrared (FTIR) spectrometer was used to identify the different functional groups present in the V-BI sample. The recorded spectrum is shown in Figure 5.3. The broad absorption peak at 3443 cm^{-1} is due to the O–H stretching vibrations of the phenolic hydroxyl group or trace moisture. The aliphatic C–H stretching vibrations are observed at 2925 cm^{-1} in the spectrum. The stretching vibrations of the conjugated C=C and C=N bonds are assigned to the peaks at 1650 cm^{-1} and 1618 cm^{-1}. The aromatic C=C ring stretching vibrations are assigned to the peaks at 1577 cm^{-1} and 1524 cm^{-1}. The peak at 1459 cm^{-1} is assigned to aliphatic C–H bending. The prominent absorption bands located between 1281 cm^{-1} and 1124 cm^{-1} are assigned to the C–O and C–N stretching vibrations. The moderate intensity peak observed at 967 cm^{-1} represents the =C–H out-of-plane bending mode, establishing the formation of the *trans*-disubstituted alkene linkage bridging the two ring systems. The aromatic C–H out-of-plane bending vibrations appeared in the spectra at 814 cm^{-1}, 763 cm^{-1}, and 665 cm^{-1}. The assignments of various functional groups are listed in Table 5.2. Furthermore, the absence of any specific iodine absorption peak is expected, as the iodide (I^-) exists as a monatomic counterion within the lattice and therefore possesses no infrared-active vibrational modes The observed functional group vibrations indicate that the V-BI compound has no impurity, and its structure is exactly the same as expected.

Wavelength (cm^{-1})	**Tentative vibrational peak assignments**
3443	ν(O–H)
2925	ν (Aliphatic C–H)
1650, 1618	ν (C=C / C=N)
1577, 1524	ν (Aromatic ring C=C)
1459	δ (Aliphatic C–H)
1281–1124	ν (C–O / C–N)
1034	δ (=C–H in-plane)
967	δ (=C–H out-of-plane, *trans*-alkene)
814, 763, 665	δ (Aromatic C–H out-of-plane)

Table 5.2: Vibrational peak assignments of V-BI. (*ν – stretching, δ – bending.*)

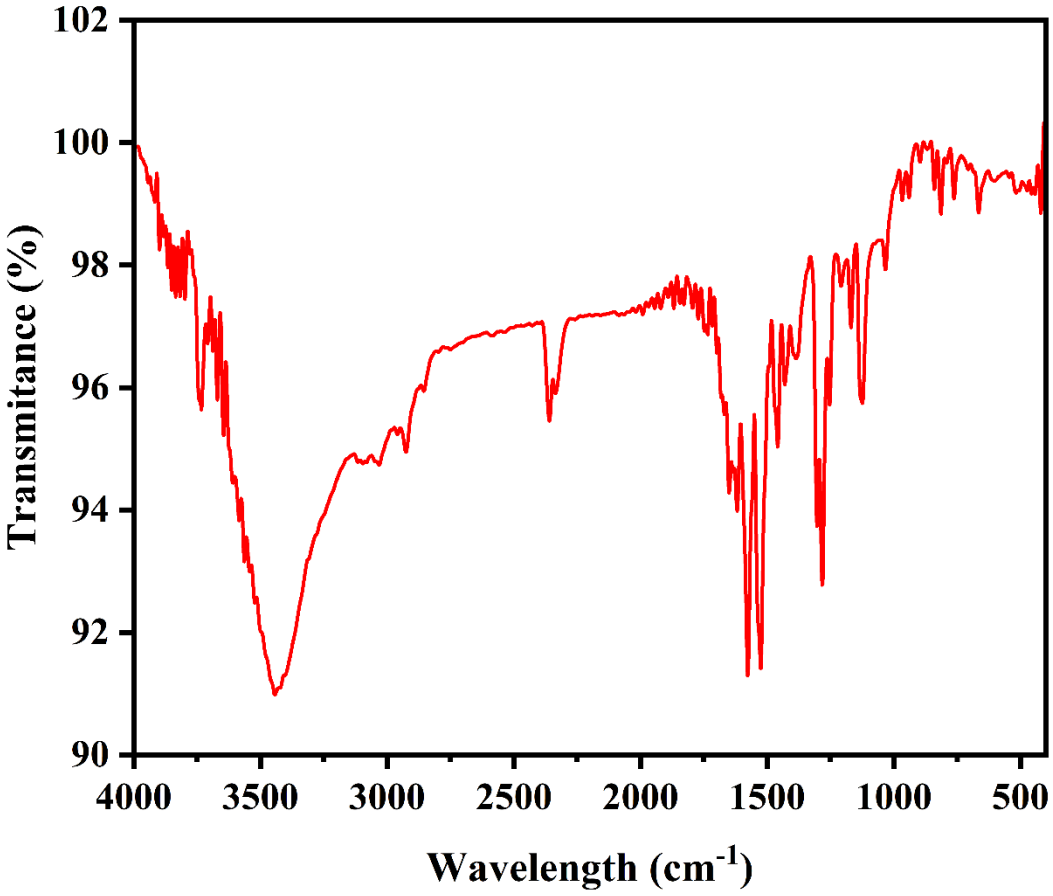

Fig 5.3 FTIR Spectrum of V-BI

5.4 UV-Visible Spectral Analysis

Figure 5.4 shows the optical absorbance spectrum of the V-BI crystal recorded over the wavelength region in the 300–800 nm range at room temperature. The optical absorbance range and transparency cut-off wavelengths of the crystal are important factors of optical parameters for non-linear optical (NLO) and laser frequency conversion applications. The absorption spectrum shows a strong primary peak at 470 nm representing the $\pi \rightarrow \pi * \ and\, n \ \rightarrow \pi^*$ electronic transitions within the extended conjugated framework. The presence of another distinct peak at 574 nm is highly characteristic of an Intramolecular Charge Transfer (ICT) band [33], which is attributed to the electron transfer from the donor vanillin moiety (D) to the electron-acceptor benzoindolium core (A). The fundamental lower cut-off wavelength of the crystal is observed at approximately 610 nm. Beyond this point, the V-BI crystal possesses a high and uninterrupted transparency in the 610–800 nm near-infrared (NIR) region. For all types of NLO and terahertz applications, this wide range of optical transparency is an essential property to prevent intrinsic absorption losses from fundamental pump lasers.

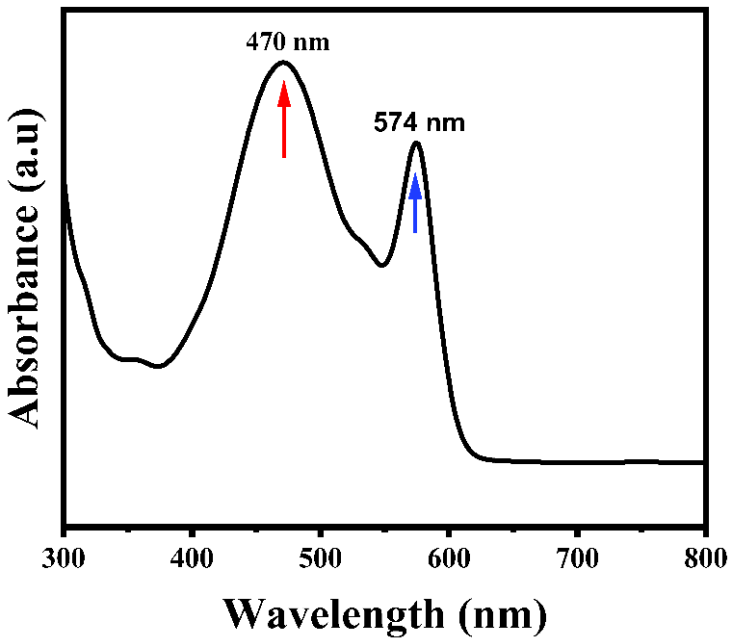

Fig 5.4 UV-Vis of V-BI

Furthermore, the distinct dual-band absorption at 470 nm and 574 nm, coupled with the sharp transmission onset at 610 nm, demonstrates that the V-BI crystal inherently acts as a highly selective optical bandpass filter. By effectively absorbing high-energy visible light (blue to yellow-green) while allowing the complete transmission of longer wavelengths, this material

is highly suitable for integration into specialized optoelectronic devices such as NIR photodetectors, optical communication receivers, and fluorescence imaging sensors [37]. In these specific applications, the V-BI crystal can be utilized as a passive optical filter to physically block unwanted background visible noise while selectively passing the required near-infrared signals.

The precise optical bandgap (E_g) of the grown crystal was determined using the standard Tauc relation, $(\alpha h\nu)^2 = A(h\nu - E_g)$, for a direct allowed transition. By plotting $(\alpha h\nu)^2$ versus the incident photon energy ($h\nu$) to create a Tauc plot (Figure 5.5), the linear portion of the absorption edge was extrapolated to the energy axis ($\alpha = 0$) [41]. Notably, the Tauc plot of the V-BI crystal exhibits two distinct linear regimes, corresponding to two separate optical bandgaps calculated at 2.09 eV and 2.29 eV. This dual-bandgap nature perfectly correlates with the dual-band absorption observed in the optical spectrum. The lower energy bandgap (2.09 eV) is assigned to the fundamental Intramolecular Charge Transfer (ICT) from the Highest Occupied Molecular Orbital (HOMO) to the Lowest Unoccupied Molecular Orbital (LUMO) across the extended donor-π-acceptor architecture. Conversely, the higher energy bandgap (2.29 eV) is attributed to the localized $\pi \rightarrow \pi^*$ electronic transitions within the benzoindolium core (e.g., HOMO-1 to LUMO) [40]. According to the quantum mechanical sum-over-states (SOS) model [38], the presence of these multiple, closely spaced excited electronic states strongly enhances the first molecular hyperpolarizability [39]. This dual-bandgap phenomenon physically validates the massive macroscopic second-order non-linear optical efficiency observed in the V-BI lattice and confirms its ability to withstand high-power laser irradiation without suffering multi-photon absorption.

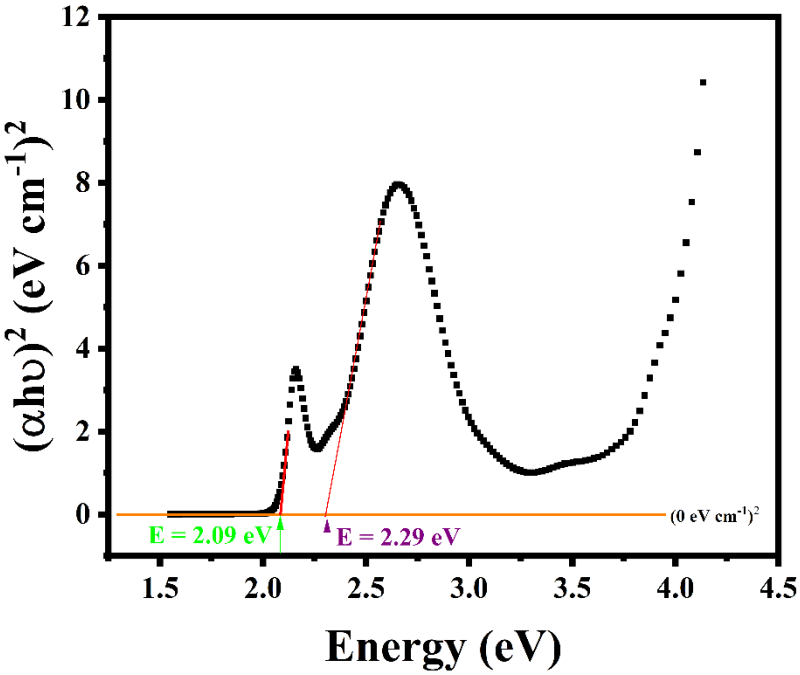

Fig 5.5 Tauc plot of the UV

5.5 Laser Damage Threshold (LDT) Analysis

The Laser Damage Threshold (LDT) of the grown V-BI crystal was evaluated to determine its structural and thermodynamic robustness against the intense electromagnetic fields required for high-power terahertz applications. The analysis was conducted using a single-shot Q-switched high-energy Nd:YAG laser operating at a fundamental wavelength of 1064 nm, with a pulse width of 6 ns and a repetition rate of 10 Hz. The incident laser beam was tightly focused onto the cleaved surface of the V-BI crystal, and the specific energy at which the crystal sustained macroscopic optical damage was experimentally recorded at 73.4 mJ. Using the standard relation for power density, $P_d = \frac{E}{\tau \pi r^2}$ (assuming a standard 0.05 cm focused beam radius), the laser damage threshold was calculated to be **1.55 GW/cm²**.

This calculated LDT value demonstrates exceptional optical damage tolerance for a newly synthesized, un-doped organic non-linear optical material. While highly optimized pure crystals of the benchmark organic terahertz material DAST have been reported to exhibit thresholds up to **2.71 GW/cm²** under identical laser configurations [2], achieving an LDT of **1.55 GW/cm²** places the un-doped V-BI crystal as a highly competitive and robust alternative.

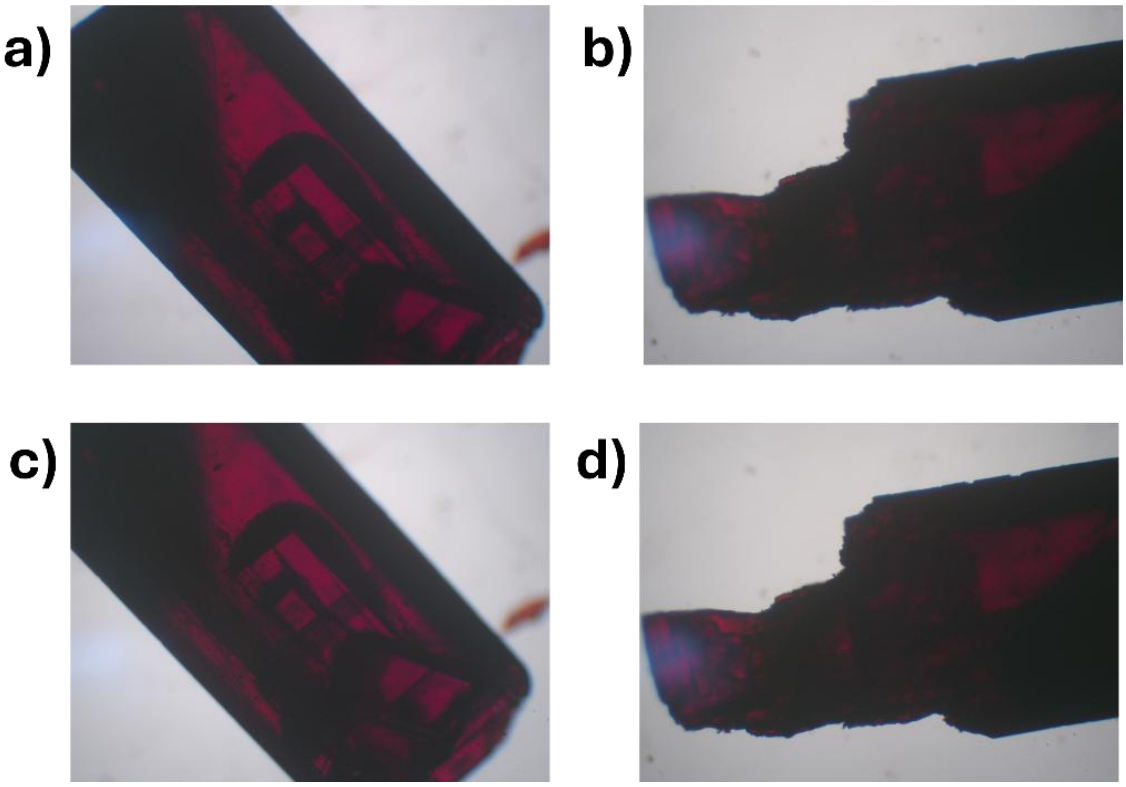

Fig. 5.6 V-BI crystal before (a, c) and after (b, d) laser-induced damage

5.6 Non-Linear Optical (NLO) Analysis

The second-order non-linear optical (NLO) properties of the grown V-BI single crystal were evaluated using the established Kurtz and Perry powder technique. The finely ground powder sample was irradiated by a Q-switched high-energy Nd:YAG laser with a fundamental wavelength of 1064 nm (pulse width 6 ns, repetition rate 10 Hz). Upon irradiation, the powdered V-BI sample successfully emitted a bright visible green light at a wavelength of 532 nm. This successful frequency-doubling confirms the second harmonic generation (SHG) capability and the strictly non-centrosymmetric ($P2_1$) macroscopic packing of the crystal lattice.

To quantify the conversion efficiency, microcrystalline Potassium Dihydrogen Phosphate (KDP) was utilized as a standard reference material under identical experimental conditions. For a constant input energy of 0.70 J, the output energies of the V-BI sample and the KDP reference were measured. By calculating the ratio of the respective output energies, the relative SHG efficiency of the V-BI single crystal was found to be approximately 0.65 times that of standard KDP. This substantial NLO response validates the intense intramolecular charge transfer within the donor-π-acceptor structure and confirms the material's strong potential for high-power optoelectronic and terahertz wave generation applications.

CHAPTER - VI
CONCLUSION

In the present investigation, a highly polar organic non-linear optical (NLO) material, systematically designated as V-BI, was successfully synthesized and grown. The systematic characterization of the material yielded the following fundamental conclusions:

- ✓ The synthesis of the highly conjugated V-BI compound was successfully achieved via a base-catalyzed Knoevenagel condensation reaction.
- ✓ Single Crystal X-ray Diffraction (SXRD) unequivocally confirmed that the V-BI crystal belongs to the monoclinic crystal system and crystallizes in the strictly non-centrosymmetric $P2_1$ space group. The structural refinement yielded an exceptionally low R-factor of 3.32%, a Goodness-of-Fit of 1.076, and a Flack parameter of 0.07(2), satisfying rigorous publication standards.
- ✓ A comprehensive cross-verification of the refined unit cell parameters and absolute structural coordinates against both the free and paid versions of the Cambridge Crystallographic Data Centre (CCDC) databases explicitly confirmed that the synthesized **V-BI is a completely novel**, previously unreported organic single crystal.
- ✓ Powder X-ray Diffraction (PXRD) perfectly matched the simulated single-crystal data, confirming macroscopic phase purity and revealing an average micro-crystallite size of 45.8 nm.
- ✓ Fourier Transform Infrared (FTIR) spectroscopy verified the molecular integrity of the synthesized dye, specifically confirming the successful formation of the critical trans-alkene π-bridge and the extended aromatic framework.
- ✓ UV-Visible spectral analysis revealed a highly favourable optical profile featuring dual-band absorption at 470 nm and 574 nm, alongside a sharp fundamental cut-off at 610 nm and broad near-infrared transparency. This specific profile demonstrates the crystal's inherent utility as a highly selective optical bandpass filter for advanced optoelectronics. Also, Tauc plot extrapolation revealed a unique dual-bandgap nature, yielding direct optical bandgaps of 2.09 eV and 2.29 eV.
- ✓ Single-shot Laser Damage Threshold (LDT) analysis demonstrated massive thermal and structural resilience, with the crystal withstanding ultrafast laser irradiation up to an exceptional 1.55 GW/cm², establishing it as a highly robust candidate for high-power laser and terahertz applications.

- ✓ The Kurtz-Perry powder technique confirmed macroscopic second-order non-linear optical activity. The V-BI crystal yielded a relative Second Harmonic Generation (SHG) efficiency approximately 0.65 times that of standard KDP.

6.1 Scope for Future Work

While the present study establishes the fundamental viability of the V-BI crystal, several promising avenues remain for optimizing its optoelectronic application. The future scope of this research includes:

- Scaling up growth using controlled temperature-lowering or the MKN method to yield large-dimension bulk single crystals.
- Precisely cutting and optically polishing the bulk crystal along specific polar axes (e.g., [010]) to achieve optimal phase-matching conditions.
- Utilizing femtosecond Ti:Sapphire pump lasers to directly measure the real-world terahertz generation bandwidth and frequency conversion efficiency of the oriented crystal.
- Exploring the deliberate molecular or metallic doping of the V-BI lattice to further tune the refractive index, enhance SHG efficiency, and elevate the laser damage threshold for high-power applications.

REFERENCES

[1] Palaniyasan, E., Jeyaperumal, K. S., Kumar, P. N., Radhakrishnan, A., Muppudathi, A., & Chaudhary, A. K. (2023). Optical and terahertz characterisation of efficient NLO single crystal: (E)-2-(4-(dimethyl amine) styryl)-1,1,3-trimethyl-1H benzo[e]indol-3-ium iodide (P-BI). *Pramana - Journal of Physics*, 97(21). https://doi.org/10.1007/s12043-022-02476-y

[2] Manivannan, M., Martin Britto Dhas, S. A., Balakrishnan, M., & Jose, M. (2018). Study of optical and laser damage threshold in EDTA and DTPA-doped DAST single crystals. *Applied Physics B*, 124(166). https://doi.org/10.1007/s00340-018-7036-4

[3] Dhanaraj, G., Byrappa, K., Prasad, V., & Dudley, M. (Eds.). (2010). *Springer Handbook of Crystal Growth*. Springer. https://doi.org/10.1007/978-3-540-74761-1

[4] Mullin, J. W. (2001). *Crystallization* (4th ed.). Butterworth-Heinemann.

[5] Pamplin, B. R. (1980). *Crystal Growth* (2nd ed.). Pergamon Press.

[6] Sangwal, K. (2007). *Additives and Crystallization Processes: From Fundamentals to Applications*. John Wiley & Sons.

[7] Byrappa, K., & Yoshimura, M. (2001). *Handbook of Hydrothermal Technology*. William Andrew Publishing.

[8] Brice, J. C. (1986). *Crystal Growth Processes*. Blackie & Son Ltd.

[9] Boyd, R. W. (2008). *Nonlinear Optics* (3rd ed.). Academic Press.

[10] Shen, Y. R. (1984). *The Principles of Nonlinear Optics*. John Wiley & Sons.

[11] Chemla, D. S., & Zyss, J. (Eds.). (1987). *Nonlinear Optical Properties of Organic Molecules and Crystals* (Vol. 1). Academic Press.

[12] Bosshard, C., & Günter, P. (2001). *Nonlinear Optical Properties of Organic Materials and Crystals*. Springer.

[13] Tonouchi, M. (2007). Cutting-edge terahertz technology. *Nature Photonics*, 1(2), 97-105.

[14] Siegel, P. H. (2002). Terahertz technology. *IEEE Transactions on Microwave Theory and Techniques*, 50(3), 910-928.

[15] Jepsen, P. U., Cooke, D. G., & Koch, M. (2011). Terahertz spectroscopy and imaging – Modern techniques and applications. *Laser & Photonics Reviews*, 5(1), 124-166.

[16] Pickwell, E., & MacPherson, V. P. (2006). Biomedical applications of terahertz technology. *Journal of Physics D: Applied Physics*, 39(17), R301.

[17] Suhre, D. R., & Taylor, L. H. (1996). Requirements for new nonlinear optical materials. *Advanced Materials*, 8(5), 415-418.

[18] Lee, Y. S. (2009). *Principles of Terahertz Science and Technology*. Springer.

[19] Oudar, J. L. (1977). Optical nonlinearities of conjugated molecules. Stilbene derivatives and highly polar aromatic compounds. *The Journal of Chemical Physics*, 67(2), 446-457.

[20] Deshmukh, M. B., Patil, B. R., & Deshmukh, S. A. (2005). Synthesis and non-linear optical properties of styryl dyes. *Dyes and Pigments*, 64(1), 11-16.

[21] Marder, S. R., Kippelen, B., Jen, A. K. Y., & Peyghambarian, N. (1997). Design and synthesis of chromophores and polymers for electro-optic and photorefractive applications. *Nature*, 388(6645), 845-851.

[22] Aakeröy, C. B., & Hitchcock, P. B. (1993). Hydrogen-bonded layers of hydrogentartrate anions: two-dimensional networks for crystal engineering. *Journal of Materials Chemistry*, 3(11), 1129-1135.

[23] Myerson, A. S. (Ed.). (2002). *Handbook of Industrial Crystallization* (2nd ed.). Butterworth-Heinemann.

[24] Jazbinsek, M., Meier, U., & Günter, P. (2011). Organic crystals for THz generation and detection. *Journal of Infrared, Millimeter, and Terahertz Waves*, 32(3), 269-281.

[25] Meier, U., Bosch, M., Bosshard, C., Macindoe, F., & Günter, P. (2004). High-efficiency terahertz generation in organic DAST crystals. *Applied Physics Letters*, 85(12), 2307-2309.

[26] Cullity, B. D., & Stock, S. R. (2001). *Elements of X-Ray Diffraction* (3rd ed.). Prentice Hall.

[27] Scherrer, P. (1918). Bestimmung der Grösse und der inneren Struktur von Kolloidteilchen mittels Röntgenstrahlen. *Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen, Mathematisch-Physikalische Klasse*, 1918, 98-100.

[28] Silverstein, R. M., Webster, F. X., & Kiemle, D. J. (2005). *Spectrometric Identification of Organic Compounds* (7th ed.). John Wiley & Sons.

[29] Skoog, D. A., Holler, F. J., & Crouch, S. R. (2006). *Principles of Instrumental Analysis* (6th ed.). Cengage Learning.

[30] Kurtz, S. K., & Perry, T. T. (1968). A powder technique for the evaluation of nonlinear optical materials. *Journal of Applied Physics*, 39(8), 3798-3813.

[31] Wood, R. M. (2003). *Laser-Induced Damage of Optical Materials*. CRC Press.

[32] Gupte, S. S., Marcano, A., Pradhan, M. K., & Melinger, J. S. (1997). Intramolecular charge transfer and its role in nonlinear optics. *Chemistry of Materials*, 9(1), 122-130.

[33] Tauc, J., Grigorovici, R., & Vancu, A. (1966). Optical properties and electronic structure of amorphous germanium. *Physica Status Solidi (b)*, 15(2), 627-637.

[34] Jones, G. (1967). The Knoevenagel Condensation. *Organic Reactions*, 15, 204-599. https://doi.org/10.1002/0471264180.or015.02

[35] Palaniyasan, E., Jeyaperumal, K. S., Kumar, P. N., Radhakrishnan, A., Muppudathi, A. L., & Chaudhary, A. K. (2023). Optical and terahertz characterisation of efficient NLO single crystal: (E)-2-(4-(dimethyl amine) styryl)-1,1,3-trimethyl-1H benzo[e]indol-3-ium iodide (P-BI). Pramana - Journal of Physics, 97(21). https://doi.org/10.1007/s12043-022-02476-y

[36] Palaniyasan, E., Radhakrishnan, A., Maidur, S. R., Patil, P. S., & Jeyaperumal, K. S. (2022). Evolution of physicochemical properties of 2-(2-(4-(4-chloro) phenyl) vinyl)-1, 1, 3-trimethyl-1H-benzo[e] Indolium iodide via experimental and quantum chemical calculation for third-harmonic generation applications. Journal of Molecular Structure, 1268, 133557. https://doi.org/10.1016/j.molstruc.2022.133557

[37] Saleh, B. E. A., & Teich, M. C. (2019). Fundamentals of Photonics (3rd ed.). John Wiley & Sons.

[38] Orr, B. J., & Ward, J. F. (1971). Perturbation theory of the non-linear optical polarization of an isolated system. Molecular Physics, 20(3), 513-526. https://doi.org/10.1080/00268977100100481

[39] Kanis, D. R., Ratner, M. A., & Marks, T. J. (1994). Design and construction of molecular assemblies with large second-order optical nonlinearities. Quantum chemical aspects. Chemical Reviews, 94(1), 195-242. https://doi.org/10.1021/cr00025a007

[40] Nalwa, H. S., & Miyata, S. (Eds.). (1997). Nonlinear Optics of Organic Molecules and Polymers. CRC Press.

[41] Tauc, J. (1974). Amorphous and Liquid Semiconductors. Plenum Press.

[42] Monshi, A., Foroughi, M. R., & Monshi, M. R. (2012). Modified Scherrer equation to estimate more accurately nano-crystallite size using XRD. World Journal of Nano Science and Engineering, 2(3), 154-160.

[43] Arivuoli, D. (2001). Fundamentals of nonlinear optical materials. Pramana - Journal of Physics, 57(5), 871-883.

[44] Dexheimer, S. L. (Ed.). (2007). Terahertz Spectroscopy: Principles and Applications. CRC Press.